INTERNATIONAL
RAILWAY SOLUTION

IRS
90940 - Ed. 1

1st edition: June 2020
Version: Original

Digitalisation, Data, Emerging Innovations

Exchange of Data

Data exchange with Driver Advisory Systems (DAS) following the SFERA protocol



© International Union of Railways, 2020. All rights reserved.
This document may not be reproduced – even in part – without the written authorisation of UIC.

IRS to be classified in volumes of UIC:
Chapter 9 - Digitalisation, Data, Emerging Innovations
Subsection 4 - Exchange of data

Complete reference in case of citation:
IRS-90940:2020,Ed1: Digitalisation, Data, Emerging Innovations - Exchange of data - Data exchange with Driver Advisory Systems (DAS) following the SFERA protocol

Approbation date:

With effect from:

Application recommended to:
All Railway Undertakings and Infrastructure Managers adopting Driver Advisory Systems (DAS)

Keywords:
Driver Advisory Systems; DAS; Standalone DAS; S-DAS; Connected DAS; C-DAS; SFERA; data exchange; Traffic Management System; TMS; ATO over ETCS; AoE; SUBSET-126; ETCS; ERTMS; punctuality; energy efficiency; energy savings; cost savings; interoperability; timetables; capacity optimisation.

Warning

No part of this publication may be copied, reproduced or distributed by any means whatsoever, including electronic, except for private and individual use, without the express permission of the International Union of Railways (UIC). The same applies for translation, adaptation or transformation, arrangement or reproduction by any method or procedure whatsoever. The sole exceptions - noting the author's name and the source - are "analyses and brief quotations justified by the critical, argumentative, educational, scientific or informative nature of the publication into which they are incorporated".
(Articles L 122-4 and L122-5 of the French Intellectual Property Code).
© International Union of Railways (UIC) - Paris, 2020

Published by the International Union of Railways (UIC)
16 rue Jean Rey, 75015 Paris - France, June 2020
Dépôt Légal June 2020

ISBN 978-2-7461-2929-0

Record of updates:

1st edition, June 2020	Creation of the document.

CONTENTS

Foreword 1

Executive summary 3

Field of application 5

Reference documents 6

Abbreviations and Symbols 8

Concepts, terms and definitions 10

General Application

1 – Introduction 14

1.1 – DAS Environment 16

1.2 – Driver Advisory Systems 18

1.2.1 – Standalone DAS 21

1.2.2 – Connected DAS 21

1.2.3 – Automatic Train Operations (ATO) 23

2 – Requirements of the exchange protocol 24

2.1 – Introduction requirements for SFERA 24

2.2 – Interoperability requirements 24

2.3 – Evolution requirements 26

2.4 – Requirements supporting different setups 27

2.5 – Communications requirements 28

2.6 – Data integrity requirements 29

2.7 – Functional data requirements 30

2.8 – Maintainability requirements 31

3 – The Journey Profile Model 32

3.1 – Overview 32
3.2 – Description of the model 32
3.2.1 – The Segment Profile (SP) 32
3.2.2 – The Train Characteristics (TC) 33
3.2.3 – The Journey Profile (JP) 33
3.2.4 – The Status Report (SR) 34
3.3 – Application 34
3.4 – Possible setups in which to use SFERA 35
3.4.1 – IM-RU setup: "Back-office" communication between the IM systems and RU systems 36
3.4.2 – IM-Train setup: Direct communication between the IM trackside system and the on-board DAS 37
3.4.3 – AoE setup: ATO over ETCS connection 38
3.5 – Additions to ATO over ETCS 38
3.6 – DAS Status and Capabilities 39

4 – Use Cases and Prerequisites 41

4.1 – General presentation of the use cases 41
4.2 – Prerequisites 42

5 – Data Structure 43

5.1 – Messaging 43
5.2 – Message headers 44
5.3 – Message payloads 44
5.4 – Identification of JPs, TCs and SPs 45
5.5 – Error codes 45

6 – Communication Mechanisms 46

6.1 – Introduction 46
6.2 – From use cases to data exchange 47
6.3 – Layered protocol 47
6.3.1 – Presentation layer 47
6.3.2 – Authorisation layer 48
6.3.3 – Communication layer 48

6.4 – Design specifications 49
6.4.1 – Data exchange 50
6.4.2 – Authentication/authorisation 50
6.4.3 – Security/integrity 52

7 – Maintenance of SFERA 53

7.1 – The need for maintenance 53
7.2 – Procedure to update SFERA 53
7.3 – How to make solutions compatible with the latest version of SFERA 54

Appendix A: SFERA Use Cases 55

Appendix B: SFERA Communication Protocol 56

Appendix C: SFERA Implementation Cases 57

Appendix D: SFERA XML Schema Definition 58

Appendix E: SFERA Data Handbook 59

Appendix F: Sample SFERA XML Communications 60

Appendix G: Correspondence Table SUBSET-126/SFERA 61

Appendix H: Error codes in SFERA 62

Regulatory listing 63

Bibliography 64

Foreword

An International Railway Solution (IRS) is a document drawn up by consensus and applied on a voluntary basis which aims to facilitate and harmonise railway operation.

IRSs are gradually replacing UIC leaflets.

Members of the International Union of Railways (UIC) participate in the creation of IRSs, of which they benefit in full. Furthermore, marketing to railway stakeholders contributes to the sharing and harmonisation of good practices, in accordance with UIC's primary goal.

While IRSs do not have the force of law, they are created and published in a professional manner and with technical approval on the basis of UIC procedures, and have full technical legitimacy in the railway sector.

The "General application" section constitutes the heart of the IRS and contains the working group's expertise applicable in the broadest manner. The addition of a "Specific application" aims to cover factors specific to a region, a type of service, etc.

Some provisions shall be implemented in all cases to ensure correct application and maintain the system's cohesiveness. Certain operational circumstances require that specific provisions be systematically applied by all users of the document.

For the purposes of understanding, the user acknowledges that the expression of:

- obligation indicates requirements to be respected in order to comply with the expertise expressed in the document (usually "shall");
- recommendation indicates options that are favourable but not essential (usually "should");
- permission indicates options that are recognised but without consequence (usually "may");
- capacity indicates possibilities from a technical standpoint and not permissions or probabilities (usually "can").

Obligation, recommendation and capacity can be expressed in the negative to indicate prohibiting, non-recommendation and incapacity.

If applicable, notably for the purposes of compliance with a UIC approval, passages that are recommended or mandatory for the cohesiveness of the system will be accompanied by the letters "R" and "MCS" respectively.

This IRS reflects the following UIC working group expertise:

UIC SFERA Project Working Group

It integrates in full or in part the following UIC leaflets:

none

It complements the following standards:

none

The following IRSs constitute a complementary series:

none

Executive summary

See Reference documents - page 6 and Abbreviations and Symbols - page 8.

This IRS is the **outcome of collaborative work** handled by leading railway actors between 2017 and 2020 under the purview of the UIC Energy, Environment and Sustainability Platform. External stakeholders were involved in the work through a workshop and draft document sharing.

Driver Advisory Systems (DAS) are IT tools which provide **live information to the drivers** to help them arrive at stations[1] on time while reducing energy consumption. DAS take into account various types of data, such as live positioning, infrastructure characteristics, the real-time transport plan, rolling stock data, etc.

The SFERA protocol is meant as a single messaging standard for data exchange between Infrastructure Managers, Railway Undertakings and on-board devices to accommodate Automatic Train Operations (ATO) and DAS applications, working both over ETCS and Class B ATP systems.

It integrates in full or in part the following UIC leaflets:

- none

It complements the following standards:

- none

Commented table of contents

General application

Point 1 - page 14 gives a general introduction into Driver Advisory Systems (DAS) terminology, the context of their implementation, the purpose of designing a harmonised way of handling the exchange of data between trackside and on-board systems and coherence with the ATO implementation strategy.

Point 2 - page 24 sets out the list of requirements for the implementation of the SFERA protocol relating to the expected business outcomes.

Point 3 - page 32 defines the Journey Profile Model, which is the macroscopic set of data used by DAS. This includes the possible setups in which SFERA can be used and the different possibilities to specify DAS status and capabilities.

Point 4 - page 41 defines use cases which describe the main types of interaction between DAS and ground systems or external entities and the prerequisites to DAS implementation to be agreed between RU and IM.

Point 5 - page 43 gives an overview of the structure of the SFERA data messaging system.

1. For example, a punctuality analysis by TTG in Switzerland in 2017 showed an average improvement of 19% in punctuality (without DAS: 42% On-Time (0s) with maximum ± 30s. timing window/ With DAS: 50% On-Time (0s) with maximum ± 30s. timing window).

Point 6 - page 46 deals with communication mechanisms, the protocol layers and design specifications.

Point 7 - page 53 defines maintenance needs and procedures to keep the SFERA protocol up to date.

Appendices

This IRS also proposes a series of appendices which provide guidelines to people implementing SFERA to avoid the most foreseeable pitfalls.

Appendix A - page 55 provides complete details of the SFERA use cases touched on in point 4 - page 41.

Appendix B - page 56 provides an in-depth description of the SFERA Communication Protocol defined in point 6 - page 46.

Appendix C - page 57 covers Implementation Cases, i.e. a selection of real-life scenarios where the succession of use cases and the events that occur on a communication level in a particular setup is presented.

Appendix D - page 58 contains the SFERA XML Schema Definition, i.e. the content and format of the data that can be exchanged in SFERA.

Appendix E - page 59 is a handbook which gives detailed information on the SFERA data structure as well as guidelines and implementation suggestions.

Appendix F - page 60 lists a few samples of XML files compatible with the SFERA format.

Appendix G - page 61 contains a correspondence table between SUBSET-126 and the SFERA elements.

Appendix H - page 62 is a reference table for error codes in SFERA.

Field of application

This IRS is used to support the implementation of Driver Advisory Systems (DAS, Connected or Standalone), Traffic Management Systems (TMS) and ATO systems (ATO). It defines a standard for data exchange (the **SFERA protocol**) and includes:

- content and format of data exchanged between DAS and TMS,
- use cases for data exchange with DAS,
- communication architectures,
- basic functional requirements needed to ensure the compatibility of the systems.

This IRS does not specify:

- Safety requirements or processes related to safety,
  Note: Safety implications of the introduction of DAS are evaluated for each specific implementation, notably the processes generating the data on the IM side, and for the RU, the information presented on the Driver Machine Interface (DMI) or only used for calculation.
- The DMI and the way driving advice should be displayed,
- The calculation method for the advice message,
- The implementation of a specific solution,
  Note: This document only describes the interface. However, some use cases are described. RUs and IMs have different working relations in different countries and the division of responsibilities should therefore take place in the course of implementation in the specific countries.
- The scope of DAS functions to be implemented,
  Note: This IRS is designed to be applicable to the most basic versions of DAS and TMS as well as to more evolved systems, so that IMs and RUs can easily define their migration paths.
- Values and data that are specific to each implementation scope and could be commercially sensitive.

For these aspects, several sets of requirements and best practices already exist for different implementations of DAS and ATO systems. All these factors are important to get results, but the requirements in this IRS only apply to the exchange of data.

This document is intended:

- for Railway Undertakings and Infrastructure Managers that operate with DAS (Connected DAS in particular) or that are planning to work with it in the future;
- for DAS suppliers, in order to ease the development and implementation of their products;
- for other IT system and software providers who offer DAS-related services, to help them in defining the interface with DAS.

Reference documents

The following documents are cited in this IRS or strongly support its requirements and recommendations:

[1] *Directive 2012/34/EU of the European Parliament and of The Council establishing a single European railway area,* November 2012
https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A32012L0034

[2] *Directive (EU) 2016/797 of the European parliament and of the council of 11 May 2016 on the interoperability of the rail system within the European Union,* May 2016
https://eur-lex.europa.eu/legal-content/EN/TXT/?qid=1569250881935&uri=CELEX:32016L0797

[3] *UIC RICS Codes*, November 2012
https://uic.org/rics

[4] *ERA "List of Class B Systems": ERA/TD/2011-11 v3.0.*
https://www.era.europa.eu/sites/default/files/library/docs/recommendation/erarec-1232015_revision_of_ccs_tsi_list_class-b_systems_en.pdf

[5] *ERA Glossary of Railway Terms*, 8th November 2010

[6] *ATO OVER ETCS GLOSSARY - ERTMS Users Group, Document version 1.6*, 03/05/2018

[7] *ERTMS Assignment of Values to ETCS Variables. Version 1.20.*
https://www.era.europa.eu/sites/default/files/activities/docs/ertms_040001_etcs_variables_values_en.pdf

[8] *ERTMS/ETCS SUBSET-125-010 and SUBSET-126-0016: Published by X2Rail-1 - Result and Publications - D4.1 - ATO over ETCS GoA 2 Specification.*
https://projects.shift2rail.org/download.aspx?id=06adfb88-4d31-4938-ae3a-ba783a371b9b

[9] *ON-TIME project Deliverable 6.1: Specification of a driving advisory systems (DAS) data format.* http://www.ontime-project.eu/deliverables.aspx

[10] *Žarko Filipović, Elektrische Bahnen, Springer-Verlag*, 2015

[11] *JSON web token standard, IETF RFC 7519.* https://tools.ietf.org/html/rfc7519

[12] *MQTT Version 3.1.1 - OASIS Standard*, 29 October 2014
http://docs.oasis-open.org/mqtt/mqtt/v3.1.1/os/mqtt-v3.1.1-os.pdf

[13] *The WebSocket Protocol, IETF RFC 6455.* https://tools.ietf.org/html/rfc6455

[14] *The Transport Layer Security (TLS) Protocol Version 1.3, IETF RFC 8446.*
https://tools.ietf.org/html/rfc8446

[15] *ISO/IEC 20922:2016. Information technology -- Message Queuing Telemetry Transport (MQTT) v3.1.1,* 06-2016. https://www.iso.org/standard/69466.html

[16] *EN 61209:1999: Maritime navigation and radiocommunication equipment and systems. Integrated bridge systems (IBS). Operational and performance requirements, methods of testing and required test results,* November 1999

[17] *IEC 62290-1:2014: Railway applications - Urban guided transport management and command/control systems - Part 1: System principles and fundamental concepts,* July 2014

[18] *EN 13816:2002: Transportation - Logistics and services - Public passenger transport - Service quality definition, targeting and measurement,* September 2002
http://standards.cen.eu/dyn/www/f?p=204%3A110%3A0%3A%3A%3A%3AFSP_PROJECT%2CFSP_ORG_ID%3A15341%2C6301&cs=18FE10A71F5CAEA8B1BED737846DA8B81

[19] *A Universally Unique IDentifier (UUID) URN Namespace, IETF RFC 4122*
https://tools.ietf.org/html/rfc4122

Abbreviations and Symbols

AoE	ATO over ETCS
ATO	Automatic Train Operations
ATP	Automatic Train Protection
B2G	Board to Ground
C-DAS	Connected DAS
C-DAS-C	Connected DAS with Centralised intelligence
C-DAS-O	Connected DAS with On-board intelligence
DAS	Driver Advisory System
DAS-OB	DAS On Board
DAS-TS	DAS Trackside
DMI	Driver Machine Interface
EoA	End of Authority
ERTMS	European Rail Traffic Management System
ETCS	European Train Control System
G2B	Ground to Board
GoA	Grade of Automation
IM	Infrastructure Manager
IRS	International Railway Solution
JP	Journey Profile
JSON	JavaScript Object Notation
MQTT	Message Queuing Telemetry Transport
NTC	[ETCS Level] National Train Control
OBU	On-Board Unit
RU	Railway Undertaking

S-DAS	Standalone DAS
SFERA	Smart communications For Efficient Rail Activities
SP	Segment Profile
S2R	Shift2Rail
SR	Status Report
TAF	Telematic Applications Freight
TAP	Telematic Applications Passengers
TC	Train Characteristics
TCMS	Train Control and Management System
TCP/IP	Transmission Control Protocol (TCP) over the Internet Protocol (IP)
TIU	Train Interface Unit
TMS	Traffic Management System
TP	Timing Point
TPE	Train Path Envelope
TSI	Technical Specifications for Interoperability
TSO	Transmission System Operator
TSR	Temporary Speed Restriction
TTCP	TimeTable Control Point
UTC	Coordinated Universal Time
VKM	Vehicle Keeper Marking
XML	eXtensible Markup Language
XSD	XML Schema Definition

Concepts, terms and definitions

The subject of the glossary below is: vocabulary for Driver Advisory Systems (DAS).

(see Reference documents - page 6)

1. driving profile

set of advice presented to the driver during the journey to reach an optimised train run based on real-time data, the Train Path Envelope and all other applicable constraints

Note 1: The advice is computed on board by an algorithm.

Note 2: Examples of advice are "traction", "cruise", "coast" and "brake".

2. Global Navigation Satellite System (GNSS)

worldwide position, time and velocity radio determination system comprising space, ground and user segments

[EN 61209:1999]

Note 1: Examples of Global Navigation Satellite Systems are GPS and Galileo.

3. Grade of Automation (GoA)

automation level of train operation in which a train can be operated, resulting from sharing responsibility for given basic functions of train operation between operations staff and system

[IEC 62290-1:2014]

Note 1: The Grades of Automation are:
- GoA 0: on-sight train operation,
- GoA 1: non-automated train operation (including trains with DAS),
- GoA 2: semi-automated train operation (ATO with driver present),
- GoA 3: driverless train operation (staff still on board),
- GoA 4: unattended train operation.

4. interoperability

ability of a rail system to allow the safe and uninterrupted movement of trains which accomplish the required levels of performance

[2016/797/EU - Art. 2 (2)]

5. journey

scheduled movement of a vehicle along a single route

[EN 13816:2002]

6. Journey Profile (JP)

set of data which includes the list of Timing Points, references to Segment Profiles and Train Characteristics, and temporary changes to the infrastructure (e.g. TSR)

Note 1: The Journey Profile is defined in real time on the basis of the Train Path Envelope. It is sent from ground to board systems to calculate the driving profile.

7. roster

list or plan showing turns of duty or leave for individuals, groups or physical resources in an organisation

Note 1: In railway operations, rosters can be used for drivers, conductors or rolling stock.

8. Segment Profile (SP)

set of mostly static infrastructure data required by the DAS on board (DAS-OB) to compute the Operational Speed Profile

Note 1: The definition is derived from the ATO over ETCS Glossary.

9. Timing Point (TP)

location identified in the schedule of a train for which a specific time is set

Note 1: The Timing Point may be associated with an arrival time, a departure time or the passing time of a train not scheduled to stop.

Note 2: Derived from the ERA Glossary of Railway Terms.

10. time window

interval of time within which a train is requested to stop, start or pass at a location

Note 1: Time windows for Timing Points are set in the Journey Profile.

11. Traffic Management System (TMS)

non-safety-critical system which advises the control command and signalling system based on the current traffic situation and other internal or external information in order to improve the level of service

[ATO over ETCS Glossary]

Note 1: In this IRS, TMS includes all systems used by the Infrastructure Manager to manage traffic.

12. Train Characteristics (TC)

set of mostly static train data required by the DAS on board (DAS-OB) to compute the Operational Speed Profile

Note 1: The Train Characteristics can be:
- provided by the trackside via SFERA;
- provided directly by the TCMS;
- stored on the DAS-OB.

13. train path

infrastructure capacity needed to run a train between two places over a given period

[2012/34/EU - Art. 3 (27)]

14. Train Path Envelope (TPE)

set of data specifying, per location, a time window and, optionally, minimum and maximum speeds

Note 1: The Train Path Envelope is defined by the Infrastructure Manager's traffic control system/agents and is based on the scheduled timetable and on-line traffic regulation. It is used as the basis for the Journey Profile.

15. Transmission System Operator (TSO)

natural or legal person responsible for operating, ensuring the maintenance of and, if necessary, developing the transmission system in a given area and, where applicable, its interconnections with other systems, and for ensuring the long-term ability of the system to meet reasonable demands for the transmission of electricity

[2009/72/EU - Art. 2 (3) and (4)]

Note 1: Transmission means the transport of electricity on the extra high-voltage and high-voltage interconnected system with a view to its delivery to final customers or to distributors, but does not include supply.

16. virtual balise

point on the railway network for which the exact location on a Segment Profile (distance from start of segment) and the geographical location (latitude, longitude and optionally altitude) are known

Digitalisation, Data, Emerging Innovations
Exchange of Data
Data exchange with Driver Advisory Systems (DAS) following the SFERA protocol

General Application

1 – Introduction

See Reference documents - page 6 and Abbreviations and Symbols - page 8.

In order to meet the challenges of intermodal competition, the railway sector needs to enhance rail system performance as a whole. Railway Undertakings (RUs) and Infrastructure Managers (IMs) are therefore initiating the development of solutions to help trains drive optimally in their time slots, i.e. with high levels of punctuality and energy efficiency, and to ensure interoperability for running cross-border trains. **Driver Advisory Systems (DAS) are key tools in achieving a first level of significant improvement in performance** in the short term without requiring tremendous investment.

DAS feature among the most powerful ways of saving traction energy, generally estimated between 5% and 15%[1] depending on the type of system: "standalone" or "connected" (to the Traffic Management System).

The SFERA protocol described in this IRS aims to define technical guidelines to handle data exchange with DAS, facilitate the use of DAS on international journeys and enable their connection to Traffic Management Systems (TMS). The acronym SFERA stands for "Smart communications For Efficient Rail Activities".

Data exchange is not a means in itself, but a tool to attain business outcomes. In the case of DAS and ATO, direct business outcomes are energy savings, punctuality and cost savings.

Fig. 1 - page 15 shows multiple steps from the data transmitted using SFERA to the business outcome.

This IRS intends to define the requirements that help achieve cost-effective ways to implement the entire system (software, hardware and connectivity).

1. Experiments at SNCF in 2018 on intercity trains showed 13% energy savings.

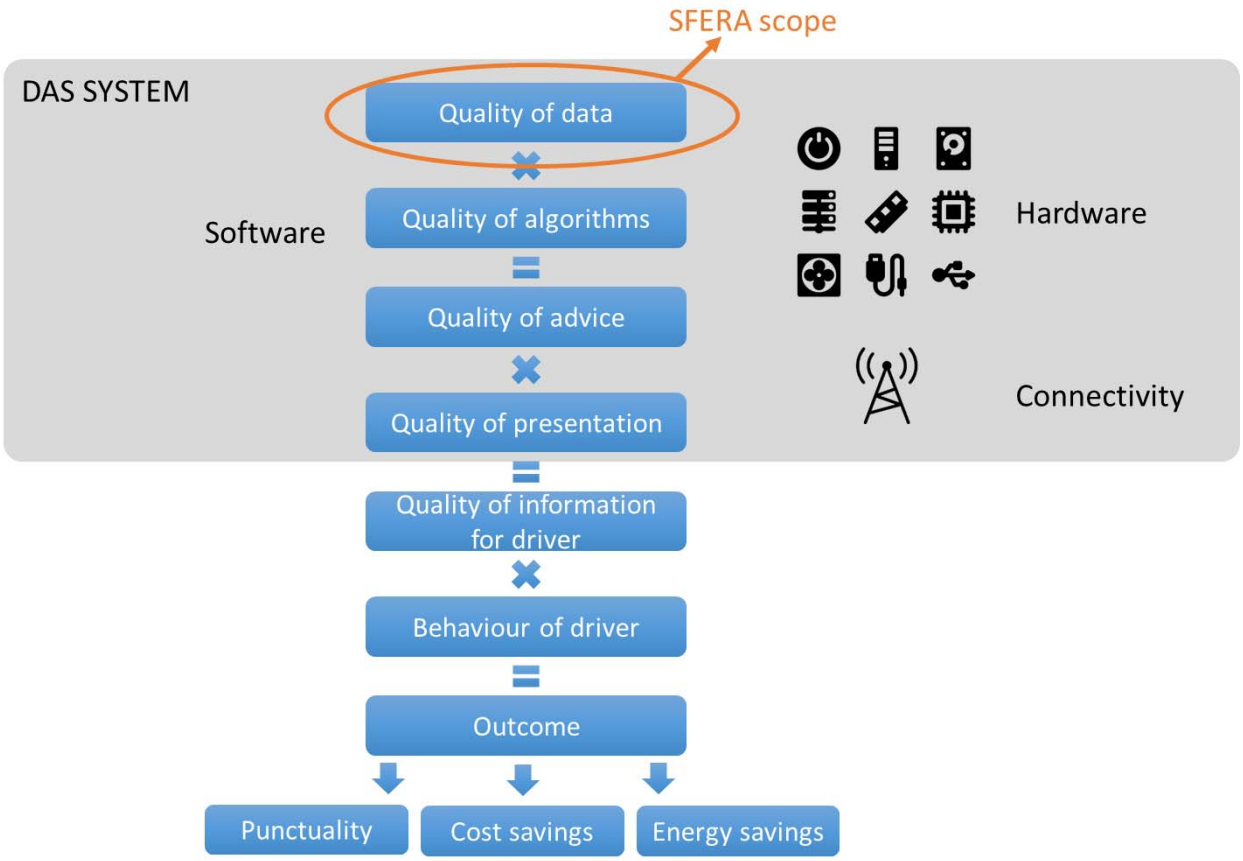


Fig. 1 - Steps from data to business outcome

The motivation for RUs and IMs to implement DAS can vary.

Indeed, these tools can contribute to energy savings, punctuality improvement, additional network capacity, power management and improved perturbation management. But the systems developed by the rail actors will have to be able to communicate with other systems. Therefore, basic guidelines need to be defined in order to allow this compatibility.

The required standard will allow the different Driver Advisory Systems in use to work across boundaries between different Infrastructure Managers and thus different Traffic Management Systems and different ATP Systems, ETCS and legacy. At the same time, the project results will also foster automation in the transmission of TMS decisions to all trains in a multi-RU environment by providing the conditions for developing "off-the-shelf" C-DAS products.

To date, some European railway companies have worked with different suppliers and implemented their own DAS able to process their own data but not the data of their "neighbours". There has been no harmonisation between systems, meaning that a DAS developed in one country cannot exchange data with the IM of neighbouring countries.

The use of a harmonised DAS protocol by all railway actors has the **following benefits**:

- RUs can choose their preferred DAS products, which will compete on functionality, efficiency and effectiveness in a fair market where players are not "locked in" to a single solution;

- RUs can increase their energy savings, as their systems will work cross-border;

- IMs can offer a single interface to all RUs, regardless of who their DAS supplier is;

- migration between products - to other Train Protection Systems or ATO systems - will be facilitated;

- better consistency in the solutions RUs and IMs invest in, which will avoid incompatibilities between systems as much as possible.

The implementation of Connected DAS is also a way to prepare longer-term implementations of ATO over ETCS by defining data exchange flows and communication systems between on-board and ground systems. The European working group on ATO over ETCS is standardising these interfaces in SUBSET-126. SFERA is defined as a superset of SUBSET-126. It is therefore possible to exchange ATO over ETCS (AoE) data with SFERA interface definition.

Every effort was made in developing this IRS to be as consistent as possible with the ATO over ETCS specifications in order to ease the transition from one configuration to the other.

Close attention was also given to setting a clear borderline between the DAS and ATP safety functions. Even if DAS indirectly has a positive impact on safety by reducing the number of restrictive signals encountered, the safety requirements and processes related to safety are handled by the ATP systems. Signal status and speed restriction information is intended to be used only for prediction purposes.

1.1 – DAS Environment

A functioning DAS environment is composed of various elements. There are components:

- on the ground, for example a Traffic Management System (TMS),

- for transmission,

- and there is equipment on the trains.

SFERA does not specify any components except the interfaces.

Fig. 2 - page 17 shows all systems in a high-level overview with their interface connections and their place on either IM's or RU's side.

The boxes on the interface indication lines point out whether the respective specifications are standardised in IRS 90940 or left to the railways to define.

IRS 90940 supports different communication setups, described in point 3.4 - page 35.

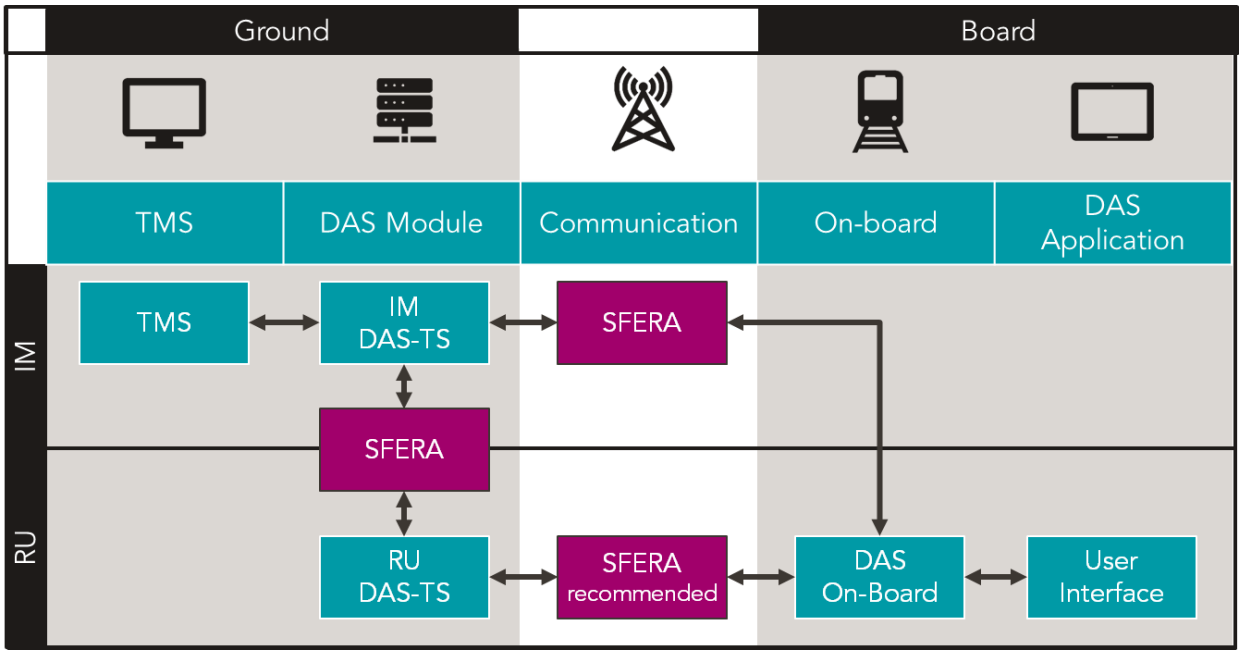


Fig. 2 - High-level system overview

Ground systems (also known as trackside systems) are:

- **TMS**,
  The TMS is operated by the IM. It serves as the source of data required to operate a DAS and calculates updated timetables for trains. The TMS forwards this information to the IM DAS-TS (usually in a specific format, either the IM's or the TMS provider's). In this IRS, the term "TMS" is used in its widest possible sense: it represents all the systems needed by the Infrastructure Manager to manage traffic. Especially included are infrastructure information - such as gradients and virtual or physical balise positions - power management functions, functions on the limitations of rolling stock (e.g. brake settings) and environment (e.g. adhesion) that may not be present in some Traffic Management Systems.

- **IM DAS-TS**,
  The IM DAS Trackside Module establishes the communication to the RU DAS-TS or directly to the DAS On-Board Units. It can be integrated into the TMS. This module creates the SFERA data packages from data received from the TMS, and transmits to or receives data from the DAS.

- **RU DAS-TS**,
  The RU DAS Trackside Module is an optional intermediate server between the IM DAS-TS and the trains' DAS On-Board Unit. Where present, it establishes connections between the ground and on-board systems. For the purpose of interoperability, it is recommended to use the SFERA protocol for data transmission between the RU DAS-TS and the DAS On-Board Unit.

Communication:

- SFERA can be used with any network connection that supports TCP/IP (e.g. Internet).

On-Board systems:

- **DAS On Board**,
  The On-Board Unit that receives the data from the ground, performs the calculation for the driving advice (if not done on the ground by RU or IM), and displays the driving advice. This device can either be integrated into the train or portable. The term DAS is frequently used instead of DAS On Board, as in this IRS.

- **User Interface**,
  The User Interface displays the driving advice to the drivers and permits data entry by a driver.

1.2 – Driver Advisory Systems

A Driver Advisory System assists drivers with the delivery of a train service. The DAS receives data from different sources and calculates the ideal driving profile for the train that fulfils the transport plan (TPE) while optimising the journey according to different criteria such as reduced energy consumption. This information is presented to the driver via an in-cab display (integrated into the rolling stock or in the form of a portable device). The DAS automatically recalculates the driving profile based on available updated information to account for late or early running compared to the timetable and then advises a revised driving profile[1].

For example, if a train is running early, the DAS will generally display a speed lower than the speed limit in order to prevent useless energy consumption and braking phases. In contrast, if the train is running late, the DAS will generally display a speed up to the speed limit, while respecting the constraints (train speed limits, environment, conflicts etc.), in order to reduce the delay.

The speed advice shown to the driver is the result of the combination of two complementary running time calculations known as the "inner/outer control loop", shown in Fig. 3 - page 19:

- "Outer loop": The **Traffic Management System establishes the optimal Train Path Envelopes (TPE)** for all trains, based on timetables, positions, Train Characteristics and estimated speeds, in order to avoid conflicts, ensure punctuality and optimise capacity usage. The TPE defines time windows for Timing Points to allow for live speed optimisation. They are sent to the DAS (or ATO).

- "Inner loop": **The DAS (or ATO) of a specific train calculates the optimal speed within the Train Path Envelopes** in order to optimise train performance. The DAS (or ATO) sends back to the TMS updated Train Characteristics, train's position and status reports (e.g. if the TPE cannot be followed), which are used for the outer loop.

1. See https://www.db-systemtechnik.de/dbst-en/news/news/pi_esg_driver_advisory_system-stagecoach_group-1679332.

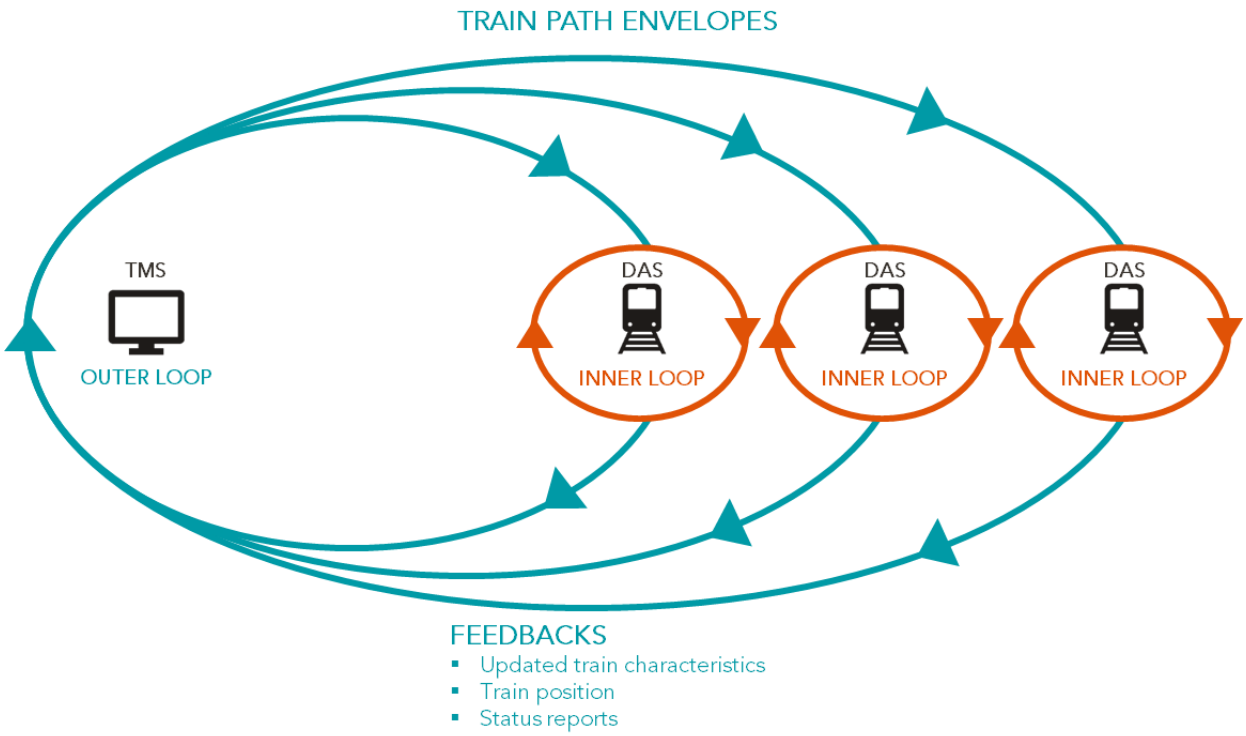


Fig. 3 - DAS Inner/Outer loop

DAS only offers advice: train drivers must make autonomous decisions and drive trains according to all the inputs they receive from the surrounding environment, as well as the legislation and rules of the country where they are driving and the instruments they have at their disposal (including but not limited to DAS).

The main goals of Connected Driver Advisory Systems are to:

- improve the smoothness and energy efficiency of a train ride;
- improve the punctuality of trains;
- avoid timetable conflicts by getting trains to arrive at the correct point in time;
- reduce maintenance costs by saving wear and tear thanks to less adverse signals, thus fewer braking phases;
- optimise the use of capacity and/or increase the regularity of a network by increasing the predictability of train runs;
- improve safety by limiting the number of restrictive signals encountered, and therefore reducing the likelihood of Signals Passed At Danger (SPAD). Lowering stress levels by using DAS also has positive effects.

Additional objectives for DAS should be to provide drivers with important information on energy supply (e.g. power limitations) and adhesion, as both factors can consequently influence a journey. The SFERA protocol supports both use cases.

SFERA does not define how the advice is displayed to the driver. Therefore, each Railway Undertaking may decide upon the design of the interface (Driver Machine Interface, DMI), taking into account national rules and regulations.

Examples of DMIs are shown in Fig. 4 and Fig. 5.

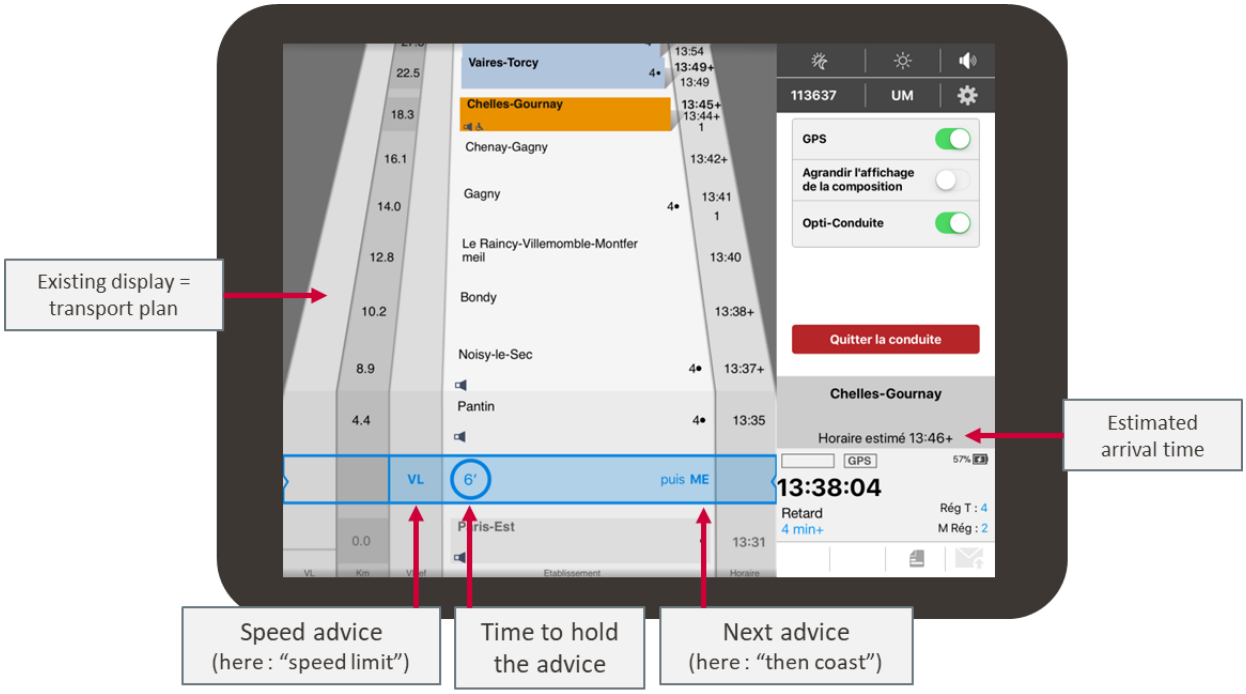


Fig. 4 - Opticonduite, "Standalone DAS" in use at SNCF

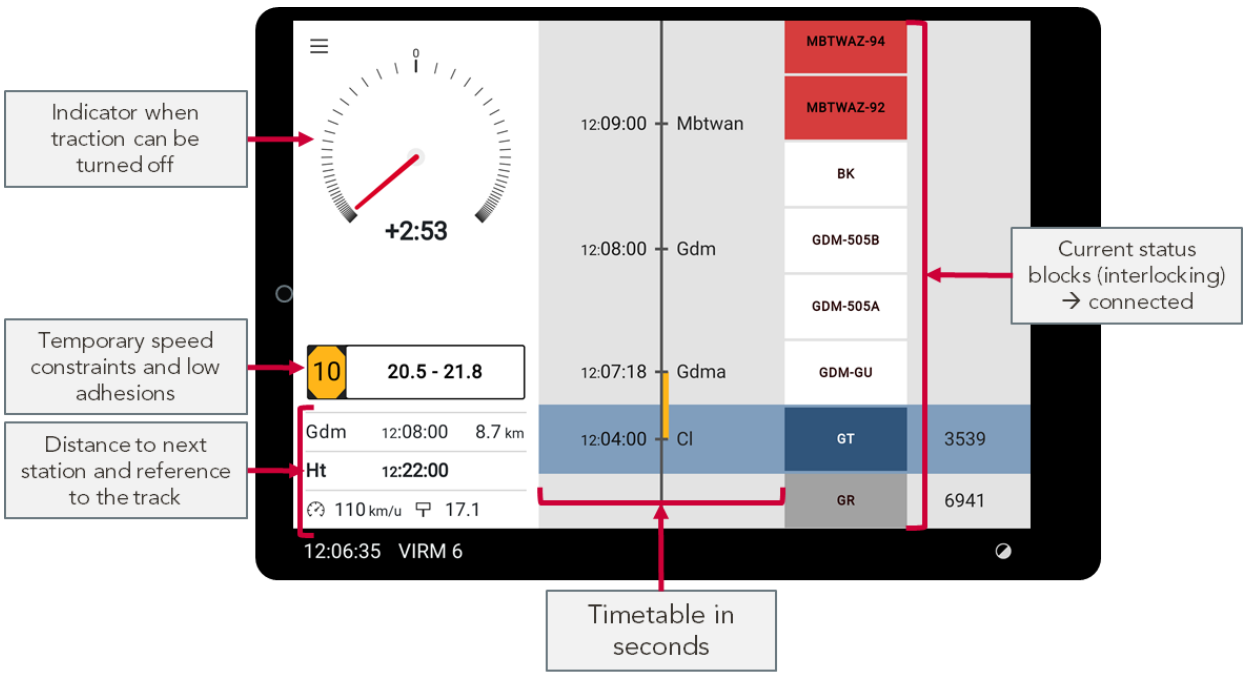


Fig. 5 - TimTim, "Connected DAS" in use at NS

Driver Advisory Systems have been slowly implemented in Europe over the last ten years (in Sweden, Germany, Switzerland, France, Great Britain, etc.), but mostly in "Standalone" versions. Most "Connected DAS", meaning DAS able to exchange data in real time with the Traffic Management System used by the Infrastructure Manager, are still in their early stages.

So far, most existing DAS are not interoperable, as they have been developed to handle a specific local data structure. This is even truer for Connected DAS, which will need to be able to switch from one data exchange flow to another as they cross borders.

1.2.1 – Standalone DAS

A Standalone DAS (S-DAS) has no connection to Traffic Management Systems while driving, but it has an information source for the position of a train (e.g. GNSS). The S-DAS is pre-loaded with static information describing the train journey, such as:

- timetable data, the first target of a DAS being to respect Timing Points;

- characteristics of the train undertaking the trip (type, length, weight, engines, maximum speed etc.). These data can be updated by the driver with "real" data;

- characteristics of the tracks (gradients, curves, signals, tunnels etc.). These attributes vary throughout the trip, so the journey will be divided into segments and the characteristics will be defined for each segment.

During the trip, the S-DAS generates speed advice for the train drivers according to these pre-loaded data, the position of the train and the current time. The driver can then get information on its position on the journey, upcoming notable points, time differentials (forecast advance or delay compared to the timetable) and optimal speed to maintain a smooth, efficient and punctual service.

The S-DAS works by getting trains as close as possible to the timetable. This is mainly achieved by using existing buffers to dampen smaller delays, thus reducing escalation to disturbed situations. The S-DAS is not able to provide data in a disrupted situation, as it does not receive live information from the environment. But it can help to recover the normal situation after a disruption.

1.2.2 – Connected DAS

The Connected DAS (C-DAS) is an evolution of S-DAS which has a connection to the TMS in order to give and receive updates about a train's journey. The C-DAS receives updates from the TMS in order to maintain a constantly updated understanding of the environment relating to the trip (e.g. conflicts, new timetables etc.), and in turn may keep the TMS informed on the status of the train journey.

The DAS constantly recalculates and displays a conflict-free and energy-optimised driving profile from real-time information about a train's path, infrastructure constraints, topology, timing targets, etc.

There are three different alternatives in the Connected DAS architecture. The definitions were established by the ON-TIME project (N.B.: the ON-TIME project[1] used the term "trajectory". In this IRS, for glossary coherence purposes, the term "trajectory" has been replaced by "driving profile"):

- C-DAS-C (Central): "[Driving profile] and advice are computed centrally. The advice is sent to the On-Board Unit (OBU) for display purposes [only]".

- C-DAS-I (Intermediate): "A proposed [driving profile] is computed centrally and transmitted to the OBU. In the OBU, the advice is computed and displayed".

- C-DAS-O (On Board): "The [computed Train Path Envelope is transmitted] to the OBU. Here, the optimal [driving profile] is computed, and the advice is calculated according to the [driving profile] and displayed to the driver".

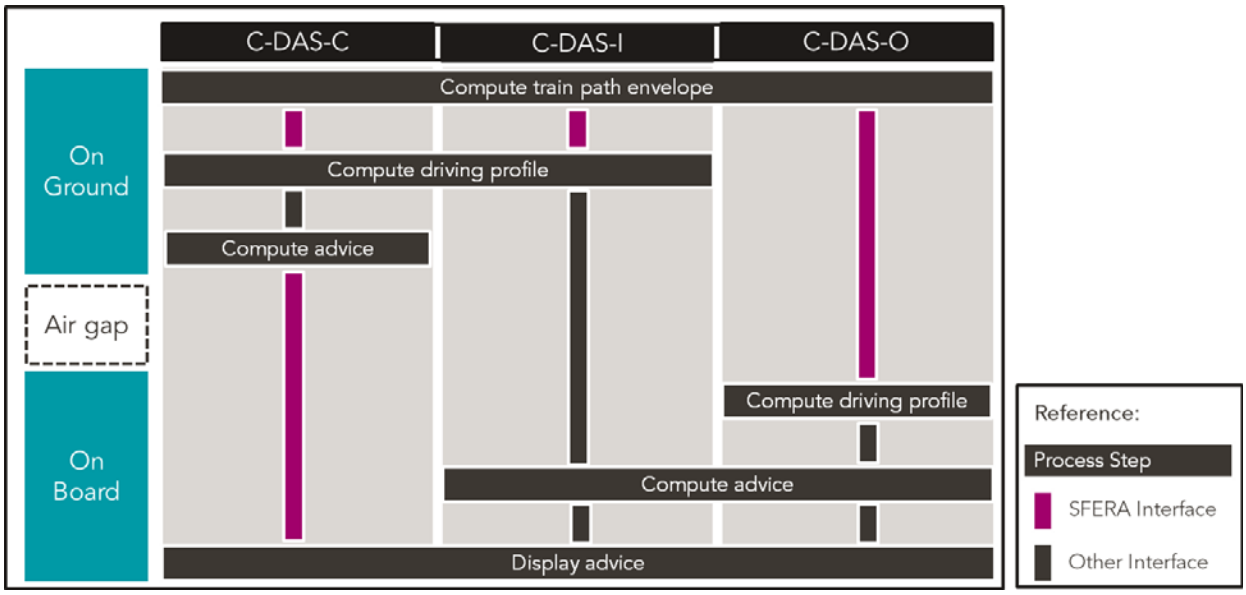


Fig. 6 - Possible C-DAS architectures

The C-DAS architectures are displayed in Fig. 6. Note that the same three interfaces are present in all three architectures. The only difference is the place in the overall process where communication between trackside and train takes place. This is often a matter of convenience (reliability and maintainability of the computation and communication devices), price (costs for installation, on-board devices, etc.) and expected performance (more or less latency due to transmission quality according to the location for the calculation).

This IRS focuses mainly on the **C-DAS-O** approach because this is also pursued in the ATO over ETCS model. This choice was made to have both protocols consistent with each other. The principal method of C-DAS operation considered here is therefore the transmission of the Train Path Envelope from the ground to the train, where driving profile and driving advice are calculated and displayed.

1. See http://www.ontime-project.eu/.

However, to allow for the railways which operate under C-DAS-C to transition smoothly to C-DAS-O, the SFERA model was equipped with additional data elements to support **C-DAS-C**. This enables IMs to calculate driving advice locally in the TMS and to send this information to the RUs or directly to the trains for display. This can be used for DAS that are not able to measure position and speed. Only DAS with a behaviour setting of "C-DAS-C-ONLY" (see point 3.6 - page 39 for additional details about DAS behaviour) may display these advice messages. There are no provisions for transitions between C-DAS-C mode and other modes. None of the data from C-DAS-C will be displayed in any other mode than C-DAS-C. Localisation for C-DAS-C is provided by the IM and there are no references to Journey Profiles or Train Characteristics.

C-DAS-I is not supported in this standard.

1.2.3 – Automatic Train Operations (ATO)

The functionality of DAS and ATO is very similar: data about the infrastructure, the train and the timetable are used to calculate a driving profile. The difference in the two systems lies in this driving profile's execution. The DAS presents it in the form of driving advice to the driver, while with ATO the train automatically controls its movements accordingly.

Ongoing projects currently specify a data exchange between ground systems and automatic trains on ERTMS lines: so-called ATO over ETCS (AoE). These specifications are laid out in UNISIG SUBSETS 125 and 126. SFERA is based on these standards and all data elements in SUBSET-125/126 can therefore also be found in SFERA.

In order to use SFERA on conventional Class B ATP Lines as well, additional data had to be added to the protocol. These are mainly elements about the infrastructure used for localisation and information about the train itself - data that on ERTMS lines is provided by ETCS system components.

This combination of the AoE data model with additional elements guarantees the DAS functionality. It also allows the use of the SFERA protocol for ATO operations on conventional Class B ATP lines (when permitted by the national safety authority). This approach was deliberately adopted to create a basis on which SFERA can be upgraded to AoE - once a line is fitted with ERTMS.

2 – Requirements of the exchange protocol

See Reference documents - page 6 and Abbreviations and Symbols - page 8.

2.1 – Introduction requirements for SFERA

Requirements for SFERA are derived from the business outcomes (punctuality, energy and cost savings), but are only a part of how to achieve them.

The requirements for SFERA fall into different categories. These requirements are further elaborated in the following points:

- 2.2: Interoperability,
  In order to achieve energy savings and punctuality in a cost-effective manner, the interoperability of DAS and ATO solutions shall be assured by the SFERA protocol.

- 2.3: Evolution,
  To obtain results as quickly as possible with limited investments, these solutions shall be able to evolve over time.

- 2.4: Setups,
  Depending on the local situation and current systems in use, implementations in various setups shall be enabled.

- 2.5: Communication,
  The communication protocol shall enable the uncorrupted transfer of data to the appropriate parties or systems

- 2.6: Integrity,
  The SFERA protocol shall have mechanisms that ensure data authenticity and integrity.

- 2.7: Data,
  The functional requirements regarding which data is accommodated is defined by currently implemented or foreseen use cases for DAS and ATO solutions.

- 2.8: Maintainability.
  Not all future developments can be foreseen. Therefore, the protocol shall be designed with flexibility in mind, and the maintainability of the protocol shall be ensured.

2.2 – Interoperability requirements

In order to save costs and promote the widespread use of DAS or ATO implementations, the SFERA protocol shall facilitate interoperability. Interoperability enables DAS or ATO solutions to function across (IM-)borders, across different ATPs, from ATO to non-ATO areas and from TMS-supervised tracks to non-TMS-supervised tracks. No matter where trains run, they can all benefit from a DAS and/or ATO that communicates with IMs using the same data exchange protocol. Otherwise, each RU-IM combination would engineer their proprietary solution for this purpose, which would be more expensive. In a generic context shown in Fig. 7 - page 25, trains shall be able to run all of these routes using the same data exchange protocol.

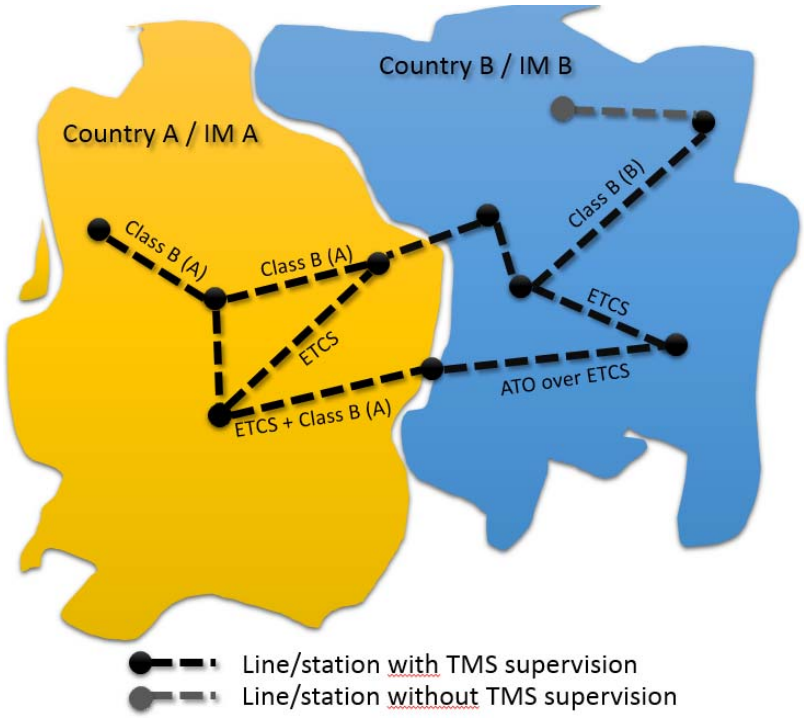


Fig. 7 - Lines with different ATPs crossing IM borders

Table 1: Interoperability requirements

ID	Requirement
IO_REQ_001	The SFERA protocol shall support the interchange of board (RU) and ground (IM) data so that DAS is able to establish driving advice for the driver.
IO_REQ_002	The SFERA protocol shall ensure the seamless transition of board data to the ground while the train moves from an area that is managed by one IM to an area that is managed by another IM.
IO_REQ_003	The SFERA protocol shall include all the data of AoE SUBSET-126.
IO_REQ_004	The SFERA protocol shall facilitate the exchange of board to ground data while the train moves from one ATP system to another.
IO_REQ_005	The SFERA protocol shall facilitate the exchange of board to ground data while the train moves from an area where it is manually operated to an area where it is automatically operated and vice versa.
IO_REQ_006	The SFERA protocol shall not implement new data interfaces for data exchange use cases/processes already covered by TSI data interfaces.
IO_REQ_007	The SFERA protocol shall be able to provide advisory information for traffic management areas and non-traffic management areas. In unmanaged areas, the DAS will operate as an S-DAS.
IO_REQ_008	The SFERA protocol shall preferably use international or widely used standards (e.g. TAP/TAF from the rail industry).
IO_REQ_009	The SFERA protocol shall be supplier-agnostic in order to achieve interoperability.

2.3 – Evolution requirements

In order to achieve cost-effective benefits for punctuality and energy savings, the quality of advice to the driver depends not only on the quality of the algorithms, but also on the quality of the data. In order to gain benefits as soon as possible, only a few elements are mandatory to start a DAS implementation. Also, it is possible to start with a less accurate set of data which improves over time. In this way, the threshold for starting is low and by using the SFERA protocol, future evolution is possible. Furthermore, the lower bandwidth requirements for less accurate data may be a reason to lower the quality on certain parts of the network.

The more data is available and the better the quality, the more precisely the DAS or ATO will be able to help trains run on time and with lower energy consumption (e.g. it shall be possible to start with a simple DAS and over time add new data elements such as gradients to the calculations) without redesigning the whole data exchange.

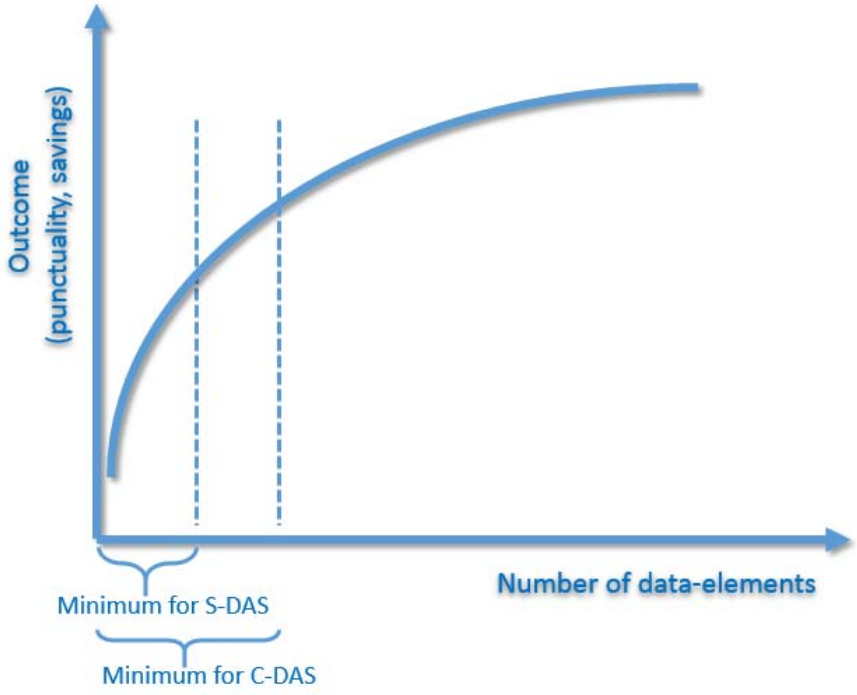


Fig. 8 - Outcome as a function of the number of data elements

This requirement, together with interoperability, necessitates indicators on data quality (precision, actuality) in the SFERA data protocol.

The same approach holds for the DAS and ATO systems: cheap implementations may use fewer data elements, even though more data is available in some of the regions where DAS is used.

Remark: Since the SFERA protocol does not enhance data, it is important that procedures for preparing and acquiring data are in place to reduce the risk of incorrect data being used by the system to an acceptable level.

Table 2: Evolution requirements

ID	Requirement
EV_REQ_001	It will be possible to use SFERA even when not all possible data elements are used.
EV_REQ_002	Data elements that ensure the proper working of the protocol and communication are mandatory.
EV_REQ_003	SFERA shall allow only the required data to be transferred.
EV_REQ_004	The SFERA protocol shall support offline (S-DAS) and connected (C-DAS) situations.
EV_REQ_005	SFERA shall define the minimum dataset in order to let it function properly.

2.4 – Requirements supporting different setups

Since there are already different implementations and not all IMs and RUs are the same, SFERA shall facilitate the different setups C-DAS-O (default) and C-DAS-C (alternative for transition purposes), as well as the different architectural setups to exchange SFERA data between an On-board system and the ground.

Table 3: Requirements supporting different setups

ID	Requirement
SU_REQ_001	The SFERA protocol shall support the exchange of data for AoE SUBSET-126.
SU_REQ_002	SFERA shall support the different communication setups IM-RU, IM-Train and AoE (as described in point 3.4 - page 35).
SU_REQ_003	SFERA shall support bi-directional communication for IM DAS-TS to RU DAS-TS, for IM DAS-TS to DAS-OB and for RU DAS-TS to DAS-OB.
SU_REQ_004	SFERA shall not interfere or conflict with other technologies such as ATO or ERTMS.
SU_REQ_005	SFERA shall enable S-DAS, C-DAS-O and C-DAS-C solutions (as described in point 1.1 - page 16).

NB: although not every country/IM is obliged to support all the different setups, it is important to keep interoperability in mind in the decision on which one will be supported. IMs and RUs will make bilateral agreements on which setup is implemented.

2.5 – Communications requirements

The communication protocol shall enable a cost-effective and swift transfer of data, ensure that the data is delivered uncorrupted and reveal information only to designated parties while trains are moving and crossing (IM) borders.

Table 4: Communications requirements

ID	Requirement
CP_REQ_001	SFERA communication shall be fair and inclusive for both small and large RUs.
CP_REQ_002	SFERA communication shall be interoperable over IM borders.
CP_REQ_003	The delivery of SFERA communication between ground and on-board systems shall be as fast as possible (within the limits of the network latency).
CP_REQ_004	SFERA communication between the ground and On-board system shall be encrypted.
CP_REQ_005	SFERA message delivery shall be guaranteed in both directions, also over shorter or longer periods of loss of physical connectivity.
CP_REQ_006	SFERA communication shall transfer complete data objects instead of only differentials.
CP_REQ_007	SFERA communication shall enable DAS to work with the last Journey Profile received.
CP_REQ_008	SFERA communication shall work on any TCP/IP-based network and can be based on public Internet.
CP_REQ_009	SFERA communication shall not be accessible to the general public, but only to specific RUs (or to entities contracted by RUs) responsible for a certain train.
CP_REQ_010	For the validity of board to ground messages, the SFERA protocol shall provide a means of authorisation of the driver actually driving on hand-held devices or on the front cab instrumentation in order to start communication (e.g. by means of position).
CP_REQ_011	SFERA communication shall allow multiple DAS instances to be connected for the same train, but only one will be active and able to send messages that have a functional impact on the TMS functions.
CP_REQ_012	SFERA communication shall be platform-independent and vendor-independent.
CP_REQ_013	SFERA communication shall fit in with the current separation of responsibilities between RUs and IMs regarding the train operation processes.
CP_REQ_014	Authentication and access management shall be independent from the rest of the SFERA protocol. The IM is not required to support these functions.
CP_REQ_015	SFERA communication shall be based on RUs being responsible for identity/access management for their own devices (including DAS devices) and personnel.

Table 4: Communications requirements

ID	Requirement
CP_REQ_016	SFERA shall not impose on RUs a specific method for dealing with identity/access management.
CP_REQ_017	SFERA communication shall work properly at all times, independently of time zone and system time differences between the communicating systems.

2.6 – Data integrity requirements

In order to guarantee that DAS solutions operate well, there shall be mechanisms in the SFERA protocol to ensure data authenticity and integrity.

Table 5: Data integrity requirements

ID	Requirement
DI_REQ_001	The SFERA protocol shall provide mechanisms to ensure that information is exchanged between the designated systems (see communication requirements).
DI_REQ_002	The SFERA protocol shall provide a means of acknowledging receipt of messages.
DI_REQ_003	The SFERA protocol shall provide a means of verifying the validity of messages (e.g. versioning).
DI_REQ_004	The SFERA protocol shall provide solutions for resending information if the data does not arrive correctly.
DI_REQ_005	The SFERA protocol shall ensure that information is not corrupted during transmission.

2.7 – Functional data requirements

In order to compute accurate advice for the driver during a train journey, DAS needs information regarding the journey (defining when the train has to be at a certain point), the features of the train which influence how fast it accelerates and decelerates and the features of the track that influence or limit the speed that can be driven.

In order to reduce power peaks (and therefore reduce investments in the electricity networks) and also to keep the trains running in case of failure of components in the power supply, or to prevent an outage of the power supply, SFERA also accommodates the transfer of data regarding use and limitation of power.

The required data is defined by the use cases of implemented or foreseen DAS and ATO solutions.

Table 6: Functional requirements

ID	Requirement
DA_REQ_001	The SFERA protocol shall have a logical data structure that allows for both mandatory and optional data.
DA_REQ_002	The SFERA protocol shall accommodate elements that influence the running of the train (e.g. the path that has to be taken by the train, the Timing Points through which the train has to pass, temporary (speed) constraints, etc.
DA_REQ_003	The SFERA protocol shall accommodate features of the train (e.g. length, weight, engine power).
DA_REQ_004	The SFERA protocol shall provide "static" data on a segment of the track (e.g. gradients, curves, static speed profile, permanent signals, tunnels) that influence acceleration and deceleration.
DA_REQ_005	The SFERA protocol shall provide the transfer of data on power (e.g. request to limit consumption or feedback with expected offtake/regeneration).
DA_REQ_006	SFERA shall provide the transfer of data on external circumstances that influence acceleration and deceleration, such as weather conditions.
DA_REQ_007	The SFERA protocol shall support the transfer of data to cope with conditions of low adhesion.
DA_REQ_008	The SFERA protocol shall support the transfer of pre-calculated speed advice (for setup C-DAS-C).

2.8 – Maintainability requirements

The SFERA protocol is based on best practices for current implementations of driver advisory systems and use cases foreseen for the near future. Over time, new functions will be needed, new versions of AoE SUBSETS will be published and therefore the SFERA protocol will need to be maintained. It is not intended to use extended SFERA messages. In order to meet the new challenges of IMs and RUs and at the same time secure interoperability and cost savings through standardisation, SFERA shall include the appropriate procedures.

In order to stay flexible and interoperable over time, SFERA shall be designed in such a manner that flexibility is more important than the minimisation of data, although for quick transfer the messages should be as small as possible.

Table 7: Maintainability requirements

ID	Requirement
MA_REQ_001	For maintenance of the SFERA protocol, procedures shall be established that provide periodic controlled updates.
MA_REQ_002	For maintenance of the SFERA protocol, procedures shall be established in order to ensure that SFERA stays compatible with AoE SUBSET-126.
MA_REQ_003	Implementations of SFERA shall be able to deal with minor updates and major updates of SFERA.
MA_REQ_004	The SFERA protocol shall, for flexibility reasons, accommodate free text messages to be sent from ground to board (G2B) and board to ground (B2G) within its structure to account for unforeseen specific use cases not covered by the "structured" part of the protocol.
MA_REQ_005	SFERA shall be easy to extend with new data elements in future versions in the maintenance phase of this IRS.
MA_REQ_006	SFERA messages shall be easy to validate against the standard.

3 – The Journey Profile Model

See Reference documents - page 6 and Abbreviations and Symbols - page 8.

3.1 – Overview

To calculate accurate speed advice during a train journey, the on-board DAS needs to be in possession of information about the infrastructure and the topology, the timetable and the train itself. In SFERA, this information - sent from the ground to the on-board system - is grouped into three categories:

- Segment Profile (SP): "static" data of a segment of the infrastructure.
- Train Characteristics (TC): the train features.
- Journey Profile (JP): "dynamic" data about the train journey, especially timings.

On the other hand, the DAS On Board needs to be able to send information to the trackside concerning the status of the train journey. This is done through "Status Reports" (SR), sent from board to the ground.

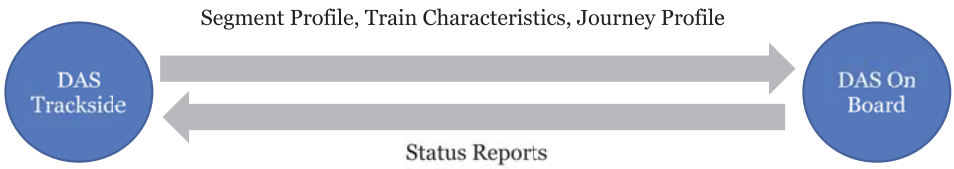


Fig. 9 - Journey Profile Model - Information Exchange in SFERA

This model of information exchange, shown in Fig. 9, is called "Journey Profile Model" in this IRS, and is derived from the ATO over ETCS standardisation (UNISIG SUBSET-125 and SUBSET-126). SFERA is mostly identical to SUBSET-125/126 to make it easy for RUs and IMs to use. Due to differences in the technical conditions which characterise ETCS and Class B ATP systems however, additional data elements had to be included in this standard to ensure functionality on legacy lines. Also, the data exchange in SFERA between the ground and on-board system happens through transmission of XML messages.

3.2 – Description of the model

3.2.1 – The Segment Profile (SP)

The Segment Profile contains the mostly static infrastructure data and represents a physical segment of the line, where the infrastructure features are usually not subject to constant change. This category includes the following data:

- Timing point locations for which a passing or stopping time may be specified in the Journey Profile,
- Notable points on the infrastructure (e.g. signals, balises and kilometre reference points etc.) and additional information and constraints regarding these elements (e.g. signal speed restrictions),

- Description of certain areas along the path of a train (e.g. platforms, tunnels or powerless sections),

- Characteristics of the infrastructure and the topology (e.g. maximum speed profiles, curves, gradients, maximum current limitations, etc.).

3.2.2 – The Train Characteristics (TC)

The Train Characteristics contain information about a train's features to calculate its dynamic capabilities. Like the SP, the TCs are mostly static since the rolling stock does not change frequently. It might seem unnecessary to send data about the train itself from the ground to the train. However, this is required because a DAS does not necessarily have to be integrated into a cabin but can also be a hand-held device with no direct connection to the train and its technical data. Furthermore, it is essential that ground and on-board systems rely on the same data for calculating running times. The following train features are included, among others:

- Information about the rolling stock type of a train, the number of locomotives and wagons it carries and about its traction and braking power.

- Physical information about the train, such as its length and weight.

- The ATP systems supported by the train.

3.2.3 – The Journey Profile (JP)

The Journey Profile is the unifying element which references both the SP and the TC. It is the most dynamic of the three data categories. It describes the unique journey (timetable) of a train on the network of a specific Infrastructure Manager. In cross-IM operations, the JPs of both data providers must be merged by the DAS. Mainly, the JP contains the following information:

- unique train identification for each train (see point 5.4 - page 45 for more details);

- timing restrictions that define the limitations the train should move within, including time tolerances and stopping point information;

- dynamic infrastructure data such as temporary constraints (e.g. low adhesion conditions or Temporary Speed Restrictions) and dynamic information relating to the ATP system a line is equipped with;

- references to Timing Point locations defined in the Segment Profiles and references to the Train Characteristics.

3.2.4 – The Status Report (SR)

The Status Report is the way in which the DAS-OB unit sends information to the DAS-TS concerning the position of the train, or data related to the current conditions of the train. For instance, the SR may be used to send the following information:

- the position and speed of the train;
- the most recent Timing Point passed or a forecast on when the next Timing Points will be reached;
- what JP is currently being used;
- changes in the characteristics of the train;
- whether there is any change in the adhesion conditions;
- data on the expected consumption and regeneration of current in the next few minutes.

3.3 – Application

In an S-DAS setup, all data will be pre-loaded on the DAS device before the train's departure and will remain unchanged throughout the whole journey, and no Status Report messages will be possible. A C-DAS will be frequently updated when there are changes such as Temporary Speed Restrictions, changes in path or timings etc., and will also include the possibility to send SRs from the on-board system to the ground.

In the data sent to the DAS-OB, the JP is the unifying element which references both SPs and TCs, which may be multiple. Every SP and TC instance has an ID and a version number that the JP references. It is possible for a train journey to be composed of multiple JPs. For example, when a train's designation changes at some point of the journey or when a train crosses a border into another IM's territory.

Generally, JPs change more frequently than SPs and TCs. This is the case because the timetable information is dependent on the operational situation, while a train's path and especially the rolling stock often remain fixed during a train journey. When a train path or the rolling stock is changed (e.g. in case of re-routings or when units are coupled during a journey), a new set of data is created with an updated ID number. If the DAS encounters a new SP or TC ID number or version in a JP, it will request the new data from the Traffic Management System (TMS) and recalculate the driving advice.

To limit transmission size, the more dynamic JP contains only a small amount of data. The larger SP and TC data sets can be pre-loaded and stored locally on the DAS On Board.

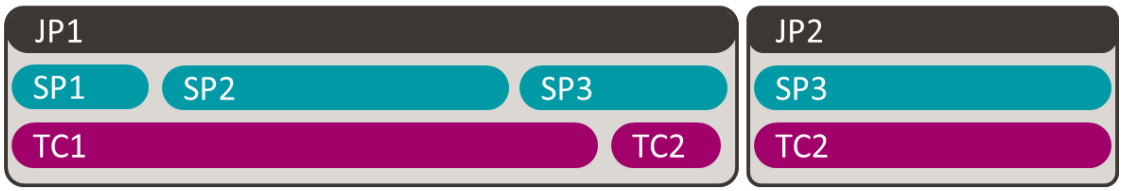


Fig. 10 - Data architecture of a sample train journey

The example in Fig. 10 shows the data architecture of a train journey with two JPs and multiple SPs and TCs.

3.4 – Possible setups in which to use SFERA

This point describes the possible architectural setups in which SFERA can be used, and the mechanisms to exchange SFERA data between the systems on the ground and the DAS on board a train.

There are three possible setups in which the SFERA protocol can be applied. These setups are:

- **IM-RU**: In this setup, SFERA shall be used as a communication protocol between IM systems and RU systems. The RU system in turn communicates with the on-board DAS devices. For this communication it is recommended to also use SFERA.

- **IM-Train**: In this setup, the on-board DAS device will connect directly to the IM systems. SFERA shall be used as a communication protocol for this channel.

- **AoE**: In this setup, SFERA is used as a protocol to connect the TMS to an ATO trackside. This is possible because SFERA is a superset of AoE SUBSET-126. This setup might make implementation on the IM side easier.

The actual setup to use is situation-specific. The aim for SFERA is to be compatible with all setups, but its implementation is left to the IM and RU. Therefore, although SFERA can be used in all setups, this does not mean that support for all functionalities in all setups is mandatory. The functionality that will be used and/or supported in a specific implementation needs to be defined and agreed on a case-by-case basis by the parties involved.

The setups are described in detail in the following three points.

3.4.1 – IM-RU setup: "Back-office" communication between the IM systems and RU systems

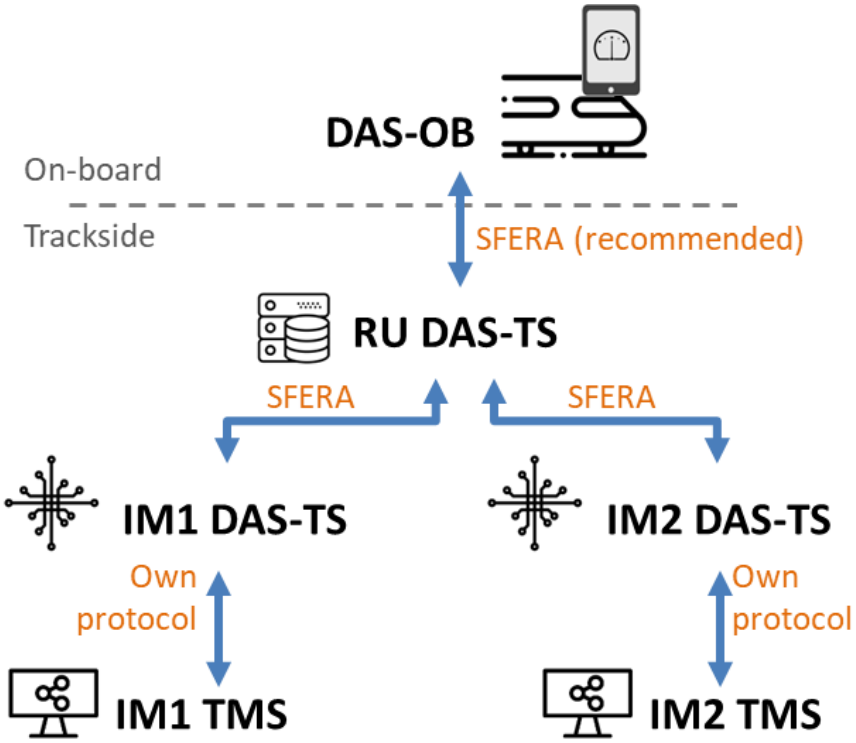


Fig. 11 - Connection through RU DAS-TS

In this setup, the IM systems will only interact with the RU systems. It will be up to the RU to act as a gateway and relay the information from the IM systems to each DAS-OB. This setup might be used if the RU wants access to the data sent to the devices for business analyses or if the IM does not want to implement a direct connection with RU devices (e.g. for computer security reasons).

The advantages of this setup are that it gives the RU influence over the devices and allows the RU to add functionality. An example use case might be that the RU itself is also interested in the information sent back from the DAS device for its own business processes.

It is important in this setup that the parties involved reach agreement, especially on the subject of information quality and guaranteed delivery.

In this case, the SFERA protocol can be found in two places: between the IM and RU, and between the RU and DAS-OB. The latter is recommended to promote interoperability.

3.4.2 – IM-Train setup: Direct communication between the IM trackside system and the on-board DAS

This setup is a more simplified version of the first setup. In this case, the on-board DAS device communicates directly with the systems of the IM that is responsible for the area where the train is running. There is no need for a ground gateway server. However, as a consequence of not having a ground gateway server, the available functionality is limited to the functionality supported by SFERA.

For example, it is not possible for the RU to get a copy of the data sent back from the device. This setup might be preferred, for instance, by RUs that have a smaller fleet, or by IMs that favour minimum latency.

This setup might also be used for applications of ATO over Class B systems.

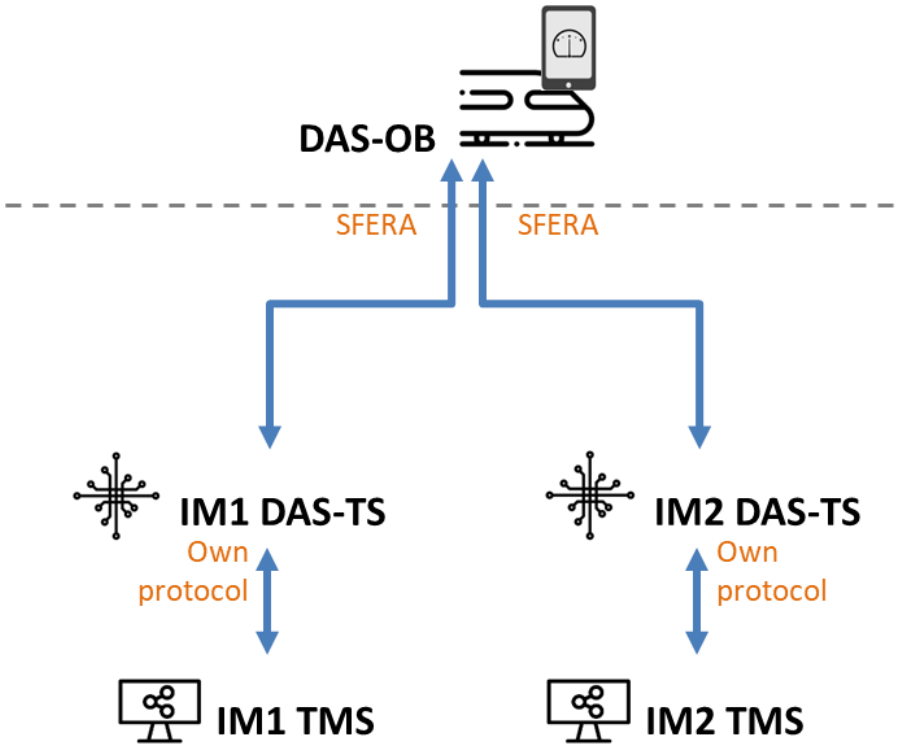


Fig. 12 - Direct connection between IM and on-board train DAS

3.4.3 – AoE setup: ATO over ETCS connection

In this setup, SFERA is used in the context of ATO over ETCS. SFERA is designed to be compatible with the standard for ATO over ETCS (specifically SUBSET-126). The information required for SUBSET-126 is only a subset of the information transferred by the SFERA protocol. As a result, it is possible to use the SFERA protocol to connect from TMS to ATO trackside equipment.

This might be interesting for an IM aiming to support both DAS and ATO over ETCS configuration. It is not required to use SFERA when connecting an ATO trackside system, but it might be a solid option.

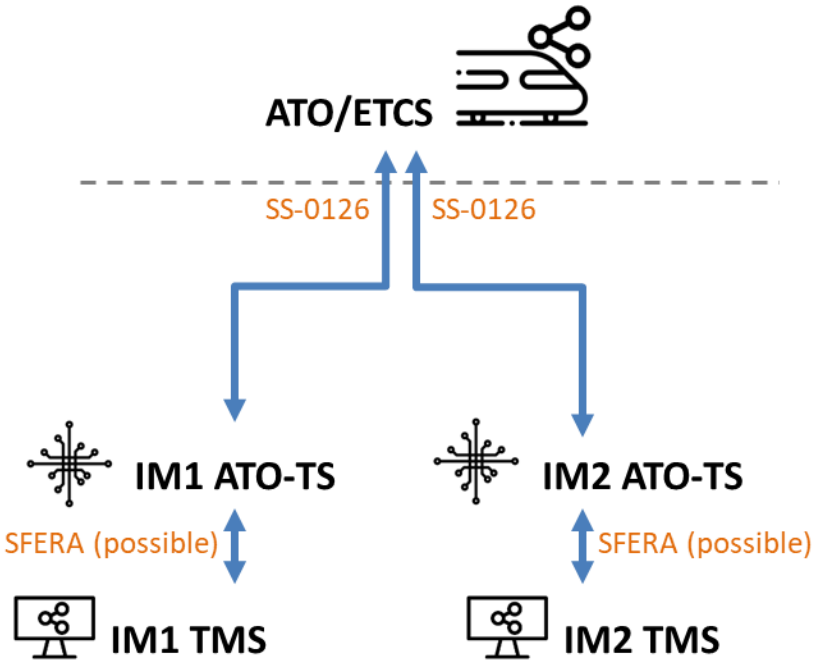


Fig. 13 - ATO over ETCS connection

3.5 – Additions to ATO over ETCS

The IRS 90940 model comes as close as possible to the ATO over ETCS standard (UNISIG SUBSET-125/126). This was intended to keep the effort involved in generating the IRS 90940 data in parallel to the UNISIG ATO data as small as possible for IMs and to offer similar data for DAS and ATO to RUs. This approach also has the advantage that the IRS 90940 data contains all the information necessary for ATO operations.

However, unlike ATO over ETCS, IRS 90940 aims to support Class B ATP lines. Due to the technical constraints of these legacy lines (e.g. the absence of ETCS balises or ETCS On-Board Units), this standard was enriched with additional data elements which on ERTMS lines would mostly be supplied by ETCS. These are mainly:

- The Train Characteristics: those used by ATO over ETCS are either received from the TCMS or hardcoded in the ATO over the ETCS On-Board Unit;

- Additional information about the infrastructure: needed by the DAS for defining a train's location (with ETCS, this information is provided by balises);

- Additional information about the signalling system, such as the position of signals, usual aspects of signals and the signal-induced behaviour of the train driver (the actual signal aspect is unknown at the time the driving advice is calculated);

- Context information about the train journey to help the driver understand the advice given (e.g. kilometre reference points, reasons associated with timetable updates).

Furthermore, while SFERA messages are exchanged in XML format, SUBSET-126 uses a binary format.

A detailed description of differences between the two models can be found in Appendix G - page 61.

3.6 – DAS Status and Capabilities

Since SFERA supports different setups as described in point 3.4 - page 35, it is essential that the ground systems are informed of the status and capabilities of the DAS-OB in order to know how to interact with it.

For instance, the on-board system may be working in Standalone DAS mode, or only be able to work with C-DAS-C, or in the case of multiple hand-held devices (e.g. for training purposes), only one should be considered held by the train driver (and therefore permitted to send data from board to the ground) while the others should be "read-only", i.e. only able to receive the G2B messages.

The DAS Status of an on-board device can be one of the following:

- *ATO GoA2*: The SFERA data is used to operate a train under ATO GoA 2. Driving profiles are used for direct train control. Status Reports will include reliable data on Train Characteristics, adhesion and power forecasts as they come from the TCMS. Text messages and C-DAS-C advice messages must be assumed to be ignored as there is no automated way to interpret them. The DAS-TS shall ensure that a maximum of one device has the status "ATO GoA2" at any time.

- *DAS Driver*: The device is operated by the driver of the train. The RU shall ensure that this is the operational driver. The position reports correspond to the actual front of the train, or a point from the front of the train associated with the distance between these two points. Changes sent to this device will be visible to the driver. The DAS-TS shall ensure that a maximum of one device has the status "DAS driver" or "ATO GoA2" at any time.

- *DAS Readonly*: The device is operated by someone who is not necessarily the driver of the train. This may be because the RU cannot ensure this, or because someone else on the train is using the device. Other users may, for example, be the next driver of the train, a trainer, or any debugging function. All advice messages shall be available. Status reporting on position, speed and power shall not be sent in the Board to Ground (B2G) Status Report. DAS-TS implementations are foreseen where either a maximum of one device can be connected (i.e. no read-only devices are allowed if one other is "DAS driver" or "ATO GoA2") or implementations where multiple DAS Readonly devices may be present.

- *Inactive*: The device is able to receive Journey Profiles but is currently not giving advice. This mode status can cover the case of an S-DAS connected to charge data for a future train service, or the case of a C-DAS that is updating data in the background, but not presenting advice to the driver.

- *C-DAS-C*: Similar to inactive, but able to show C-DAS-C advice.

DAS Behaviour can be used to inform the TMS on the capabilities of the DAS:

- *Standalone*: the DAS is operating on a Journey Profile which may have been downloaded some time earlier. The DAS may or may not process updates in JPs. Status Reports will be sent, based on the originally downloaded JPs. This status may also be delivered from the RU back office to the IM back office in the IM-RU setup in the case that the RU back office delivers cached versions with high latency to the DAS on-board devices.

- *Connected*: the DAS will respond quickly to updates to a Journey Profile. Therefore, the trackside may assume that the most recent one will be used (the DAS will send Status Reports stating the version of the JP on which the current advice to the driver is based). SRs will be sent, based on the most recent JPs.

- *C-DAS-C_Only*: Only able to show C-DAS-C advice.

It is also possible for the DAS-OB to change status or behaviour in some cases, and in specific cases, a DAS may be "forced" to change its status. The state changes between these modes are described in the DS (DAS Status) use cases, see Appendix A - page 55.

4 – Use Cases and Prerequisites

See Reference documents - page 6 and Abbreviations and Symbols - page 8.

4.1 – General presentation of the use cases

The data structure and communication layers of SFERA have been defined by taking into account a set of use cases. These use cases represent the interaction between a DAS-OB and the TMS on the ground, as well as external entities (e.g. other trains or power supply management applications), and are available to be consulted in Appendix A - page 55.

Some of the use cases described are at a process level, and others are at a data exchange level. The interpretation of the use cases may depend on the setups used (see point 3.4 - page 35 for the description of the setups).

- For those at a process level, the actors listed will be the applications that are used to perform actions on the data (e.g. TMS, DAS, RU Backoffice, Traction Power Management systems…). These use cases are directly applicable, no matter if the IM-RU or IM-Train setup is used;

- For those at a data exchange level, the systems communicating with each other are different in each setup. These use cases will refer to a communication between a receiver and a provider. The "receiver" will be the system nearest to the DAS, and the "provider" will be the system nearest to the TMS. Table 8 summarises the roles that will be taken by the different systems in the three types of links that can be encountered.

Table 8: Provider and Receiver roles in different setups

Setup	Link	Provider	Receiver
IM-RU	IM DAS-TS <-> RU DAS-TS	IM DAS-TS	RU DAS-TS
	RU DAS-TS <-> DAS-OB	RU DAS-TS	DAS-OB
IM-Train	IM DAS-TS <-> DAS-OB	IM DAS-TS	DAS-OB

In addition, sequence diagrams are described to illustrate the implementation of the use cases with the SFERA messages. For the process level use cases, the sequence diagrams are described in the context of the IM-Train Setup, which implements a push configuration. The use case appendix also presents an example of how the sequence diagrams can be extended to the IM-RU setup.

In addition to the use cases, a selection of real-life scenarios (named implementation cases) is available in Appendix C - page 57, where the following is presented:

- the succession of use cases,

- the events that occur in an IM-Train setup on a communication level (see point 5 - page 43).

Consulting that appendix could give some additional insight into understanding the links between the use cases and the interactions with the communication layer for the IM-Train setup.

4.2 – Prerequisites

Before using operationally the use cases described in Appendix A - page 55, some prerequisites must be agreed between the RU and the IM. This step has to be repeated once an RU starts using a DAS on any new IM's network, or after a major systems upgrade. This step will not be done using any specific protocol, and therefore this IRS will not impose a method. However, the following conditions are required to be met before each next step can be taken:

1. Only for the "IM-Train" Setup, the IM knows the validation mechanism for the authorisation protocol and all related data from the RU;

2. The RU knows the DNS name of the SFERA server of the IM;

3. Only for the "IM-Train" Setup, the IM and RU have validated their communication to the SFERA requirements;

4. The IM and RU have agreed on the time synchronisation accuracy;

5. The IM and RU have agreed on the mode(s) for which data is available on the IM's network (C-DAS-O and/or C-DAS-C);

6. The IM knows if Status Reports are possible from the RU's DAS;

7. The IM and RU have implemented the processes for the IM to know the TC information;

8. The RU has sufficient knowledge on the service level of the IM SFERA server;

9. The IM and RU agree that the use of the specific DAS has no negative impact while operating on the IM's network, according to the processes guiding the implementation of change on each network (e.g. Common Safety Methods);

10. The RU has implemented a process for the DAS to know the Train-IDs of the trains likely to be operated;

11. The RU has implemented a process for the DAS to know the DAS-TS to connect to in order to obtain the JP for the first part of the trip where the DAS will be used;

12. Optionally: There is an SLA for the SFERA data.

Outputs 1-4 are used in the communication protocol, and 5-12 to ensure good operation.

5 – Data Structure

See Reference documents - page 6 and Abbreviations and Symbols - page 8.

The present point provides an overview of the SFERA data structure, i.e. the actual content of the messages exchanged between on-board devices and the ground.

The complete description of the format of the XML messages that can be sent in SFERA is contained in an XML Schema Definition (XSD) available in Appendix D - page 58. This is complemented by a "Data Handbook" available in Appendix E - page 59, which gives detailed information on the SFERA data structure, as well as guidelines and implementation suggestions. Appendix F - page 60 contains sample XML communications that can be used as a basis to build workable SFERA XML files.

5.1 – Messaging

The SFERA protocol exchanges data in XML format between on-board and trackside systems. This is done through board-to-ground (B2G) and ground-to-board (G2B) messages.

G2B messages are messages going in the general direction from the IM DAS-TS to the DAS-OB, and B2G messages are those going in the opposite direction.

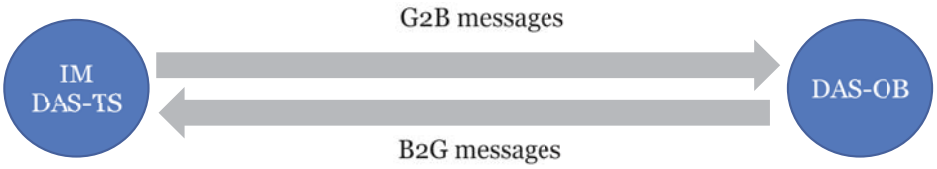


Fig. 14 - G2B vs B2G in IM-Train Setup



Fig. 15 - G2B vs B2G in IM-RU Setup

Therefore, between an IM DAS-TS and an RU DAS-TS, "Ground to Board" and "Board to Ground" messages are exchanged, although both functions are technically located on the ground.

For B2G and G2B, three types of message can be generated:

- Event messages,
  These are unsolicited messages used to communicate events.

- Request messages,
  These are messages used to request data.

- Reply messages.
  These messages are sent as a reply to either event or request messages.

Therefore, there are six types of SFERA messages in total: B2G Event/Request/Reply, and G2B Event/Request/Reply. All SFERA messages include a message header and a payload: the structure of the message header is the same for all types of messages, while the payload varies according to the type of message.

5.2 – Message headers

Each SFERA message has a message header, which contains several fields. This includes, for instance, the version of the SFERA protocol used by the message, the timestamp, the identifiers for source and destination of the message, as well as other optional data.

Each SFERA message header must contain a message ID, which uniquely identifies the message; it is assumed to be globally unique.

When a SFERA message needs to refer to earlier communications (e.g. SFERA reply messages and C-DAS-C revoke messages), the so-called "correlation ID" is used. This field can be filled with the message ID of the previous message to which the present message refers.

5.3 – Message payloads

SFERA message payloads vary according to the type of message.

- B2G Event messages are used to send Status Reports, described in point 3.2.4 - page 34.
- B2G Request messages are used to send data requests to the ground. Mostly this is used to ask for SPs, TCs and JPs (see points 3.2.1 - page 32 to 3.2.3 - page 33), but some other types of data may also be requested.
- B2G Reply messages are mostly used to send acknowledgements (OK or Errors) to ground messages, but they can also be used to send information about the DAS status and behaviour (see point 3.6 - page 39).
- G2B Event messages are mostly used to communicate updates to the Journey Profile.
- G2B Request messages are used to request the status of the DAS-OB.
- G2B Reply messages are used to respond to B2G Event or Request messages. They either contain current data (e.g. the JPs, SPs and TCs requested by the DAS-OB), or responses to the messages received, which can be positive acknowledgements or errors.

The SFERA message payloads can contain a wide range of data that provides information to be exchanged between on-board and ground systems: for an in-depth description of the data, please refer to the XSD in Appendix D - page 58 and to the Data Handbook in Appendix E - page 59.

5.4 – Identification of JPs, TCs and SPs

Segment Profiles and Train Characteristics have identifiers (IDs) and versions. These IDs and versions are referenced in Journey Profiles in order to create a journey for the train. If a JP sent to the DAS-OB references some SPs and/or TCs with IDs and/or versions that are not already loaded by the DAS-OB, the On-Board Unit requests these "new" items from the ground.

On the other hand, JPs do not have IDs: a JP is uniquely associated with a train by the Train Identification. Train Identification is used in several SFERA messages. In particular, it is used to associate a JP with a specific train.

In SFERA, it is possible to use either a TAP/TAF identifier (built following TAP/TAF specifications) or an Operational Train Number (OTN) identifier. It is preferable to use the TAP/TAF identifier. However, for transition purposes, the OTN identifier is also supported by SFERA.

However, JPs have versions. While the assumption is that the latest JP received is always correct, and the timestamp of the SFERA header should make sure that updates are not processed in the wrong order, the JP version is used to check whether the JP in use by the DAS-OB corresponds to the latest JP sent by the trackside unit.

In fact, the TMS can use the JP version as a unique identifier of the version of the JP. It is used by the DAS to send a Status Report to the trackside, communicating the JP currently used to generate advice. With the SR, if possible, the DAS will also communicate the position within a segment. These SR messages can be used by the TMS to validate that the DAS acts on the most recent information.

5.5 – Error codes

As mentioned above, SFERA Reply messages (both B2G and G2B) can contain acknowledgements to Event and Request messages. These acknowledgements can be positive (simple "OK" messages) or can contain an error.

When sending an error, the reply message can identify the location (XML Path) of the original message and provide an error code to specify the reason for the error.

The list of error codes in SFERA is available in Appendix H - page 62.

6 – Communication Mechanisms

See Reference documents - page 6 and Abbreviations and Symbols - page 8.

This point provides an overview of the SFERA communication protocol and authorisation mechanisms. Implementation specifications and details are described in Appendix B - page 56.

6.1 – Introduction

To accommodate the required flexibility in the communications and to match the separation of responsibilities between IM and RU, the protocol is designed as a set of layers. Each layer covers one or more features of the communications protocol and minimal technology choices have been made for each layer to provide interoperability.

Fig. 16 contains a functional decomposition which describes the interaction and communication between ground and board systems. Besides the exchange of data (XML), which forms the main part of this IRS, there is a layer which handles the communication between ground and board systems. This communication layer describes a bi-directional message exchange mechanism between the ground and board systems. It provides for guaranteed delivery over public Internet connections. In-between there is an authorisation layer which handles authorisation between trackside and On-Board Units and the handovers between IMs. This point contains a description of the communication layer and the authorisation layer.

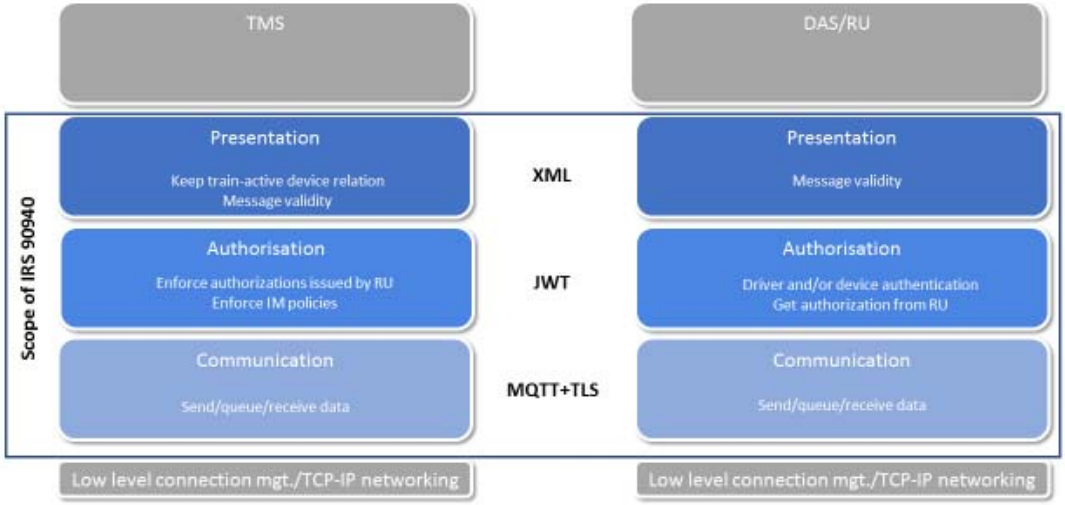


Fig. 16 - Different layers in scope of IRS 90940

The blue layers are within the scope of this IRS. However, this IRS does not specify all the behaviour in detail. In some cases (e.g. the driver/device authentication process), the decisions on the implementation details are left to the parties involved.

6.2 – From use cases to data exchange

The SFERA protocol is used in the IM-Train setup for data exchange between IM DAS-TS and RU DAS-OB, as well as in the IM-RU setup for data exchange between IM DAS-TS and RU DAS-TS. In the use cases taken into account to build the SFERA protocol, as described in point 3 - page 32 and Appendix A - page 55, four basic patterns of communication between DAS-TS and DAS-OB (either directly or through RU) can be distinguished:

1. DAS-OB requests information from DAS-TS, and DAS-TS sends a reply with the requested information (e.g. request for a Journey Profile or a Segment Profile);

2. DAS-OB sends an update of information to the DAS-TS to inform the DAS-TS of a changed situation (e.g. a position update);

3. DAS-TS sends an update message to a train (=its DAS-OB) to inform the train about a change in the system (e.g. update for a Journey Profile);

4. DAS-TS requests information about a train (=its DAS-OB) and the DAS-OB sends a reply with the requested information (e.g. a status update).

All high-level use cases are implemented with a sequence of one or more of these communication patterns. Each of these patterns is implemented as a set of interactions with the messaging system to actually transmit data.

6.3 – Layered protocol

The SFERA protocol consists of three layers to organise the different features of the protocol in a logical and flexible way (see Fig. 16 - page 46). A proper SFERA implementation (DAS, RU and IM side) should respect this organisation.

6.3.1 – Presentation layer

The presentation layer deals with XML data and consistency in exchanging the data with the TMS, independently of the transport of the data which is addressed in lower layers. The key concerns that are addressed in this layer are:

- XML validation and logical data validation procedures to ensure that corrupt data is not fed into the TMS and that corrupt data is not sent out from TMS to DAS devices. This should cover a conventional strict XML schema validation as well as additional logical value validations. Invalid messages received from the board devices should generate replies containing error messages;

- Minor version check and optional version conversions if needed for the IM DAS-TS;

- Ensuring that only messages received from the 'active' on- board device are forwarded to the IM DAS-TS;

- Handling for switching 'active' on-board device from one device to another.

6.3.2 – Authorisation layer

In the SFERA communication mechanism there is a strict separation between the responsibilities of IM and RU.

The RU owns the on-board devices and is responsible for its drivers, for the process of allocating drivers to trains and for ensuring that the intended driver is operating the intended train. An RU is also responsible for device management and hardening.

The IM controls trains, not people, and is responsible for delivering the needed information to the trains at the appropriate moments. It needs to ensure that information is distributed according to privacy laws and other regulations regarding different RUs on its infrastructure network.

This separation of responsibilities is reflected in the authorisation layer.

The technology choice in the SFERA protocol for authorisations is the use of signed JSON web tokens (*IETF RFC 7519 standard*). The reason for this choice is explained in Appendix B - page 56.

6.3.3 – Communication layer

To achieve interoperability in operations, standardisation on the technical level of exchanging data is needed. The communication layer deals with reliable communication and the logic for distributing messages to the right DAS devices so that the TMS is effectively decoupled from the actual DAS devices.

The technology choice for implementing the communication layer in the SFERA protocol is Message Queuing Telemetry Transport (MQTT) Version 5 over a secured web socket (TLS). MQTT is specified in *ISO/IEC 20922:2016*, web sockets are specified in *RFC 6455* and TLS is specified in *RFC 8446*. The reason for this choice is explained in Appendix B. The use of MQTT is dependent on the setup: it shall be used in the IM-Train setup and is recommended for the IM-RU setup.

6.4 – Design specifications

Fig. 17 shows the DAS RU/IM ecosystem components and the different setups.

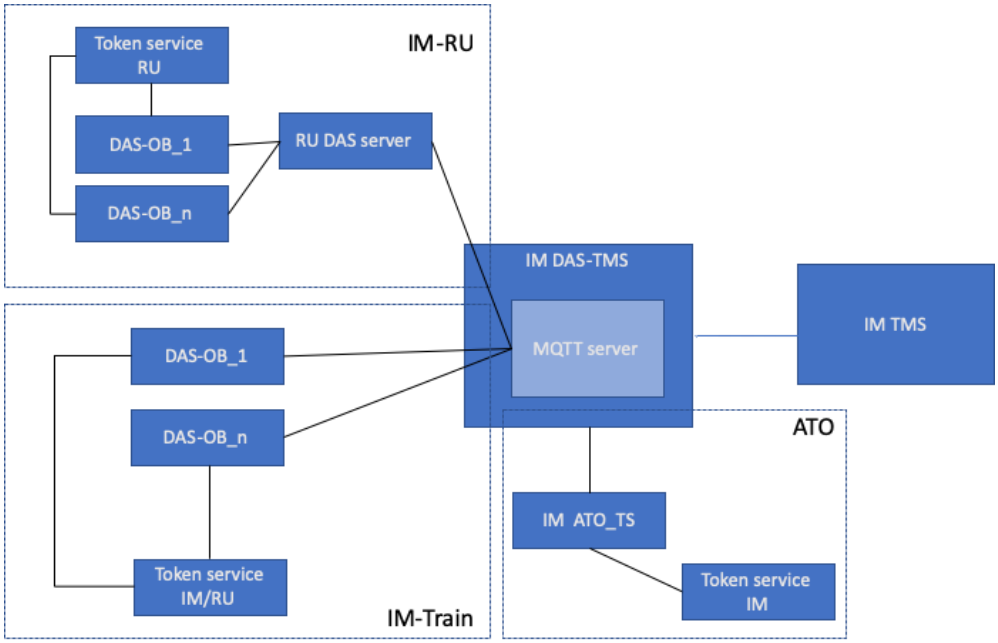


Fig. 17 - DAS RU/IM Ecosystem

The SFERA standard does not specify how the TMS integrates with the MQTT server. For the other data exchanges, the following holds:

Table 9: Design specifications for different setups

IM-Train setup	Data exchange between DAS-OB and IM DAS-TMS **shall** be implemented according to the SFERA standard (all layers).
IM-RU setup	For the IM DAS-TS to RU DAS-TS, the application layer **shall** be compliant with the SFERA XSD, and the authorisation and communication layers are subject to a bilateral agreement between RU and IM. They can decide together which layers defined in the SFERA protocol are used. The exchanges between DAS-OB and RU DAS-TS are the full responsibility of the RU. However, it is encouraged to implement the complete SFERA protocol (all layers) to facilitate the transition from the IMs that can limit their offer to only "IM-RU setup" or "IM-train setup".
ATO setup	Data exchange between IM ATO-TS and IM-DAS-TS **should** be implemented according to the SFERA standard (all layers). A completely proprietary solution is allowed.

The following points list the main design specifications.

6.4.1 – Data exchange

The data exchange specifications are listed in Table 10.

Table 10: Data exchange design specifications

Design specification	Remark
MQTT V5 is the technology to exchange data between IM and RU/DAS.	
Every IM shall make an MQTT V5 service (the SFERA service) available to provide the data for the infrastructure and train movements it is responsible for.	Service end points are communicated with contracted RUs
All communication patterns in the use cases are mapped to publish/subscribe exchanges with the SFERA service.	
A predefined topic structure shall be used in the MQTT service. This structure is organised around RUs, train movements and individual DAS devices to accommodate the one-to-many and one-to-one communications.	Refer to Appendix B - page 56 for detailed specification.
A technical connection between a DAS/RU and an IM may be disconnected and reconnected at any time.	
A technical connection between a DAS/RU and an IM may be kept open for the duration of many train movements.	
In the IM-Train setup, the SFERA service shall be accessible through public Internet connections.	
SFERA messages contain information on when to connect to another IM to get the required follow-up information.	Prepare handover
A DAS/RU may have connections to multiple IMs at the same time.	Handover

6.4.2 – Authentication/authorisation

To participate in SFERA communications, sufficient guarantees must exist that a DAS device can be trusted and is entitled to exchange information for the designated train movement.

The RU shall define a process to validate the following conditions:

- Verify the integrity of the device that is used to run the DAS application. The procedure for this depends on whether it is a built-in or handheld device;

- Verify that the user is allowed to use the DAS application;

- Verify that the user is a person designated to act on the given train (either driver or otherwise).

This process results in giving authorisation to interact with an IM and exchange information regarding a given train movement. SFERA does not specify any part of this process in detail, but relies on the agreement between RU and IM that the process is performed according to the latest security insights and current operational procedures for performing train movements. The protocol relies on the RU issuing tokens that state the authorisation of a driver operating a specific DAS, in agreement with the process above. Authorisation tokens are bound to the messages between DAS/RU and IM.

To accommodate this, the following design specifications hold:

Table 11: Authentication/authorisation design specifications

Design specification	**Remark**
A signed JSON web token (token) is used to carry authorisation information	*IETF RFC 7519 standard*.
A token contains claims for: - identification of issuing system - identification of RU - identification of DAS device - identification of SFERA service (IM) that provides the SFERA data - identification of the train movement that is to be operated - time validity of token	Refer to Appendix B - page 56 for token specification
To ensure integrity of tokens during transmission, public/private key encryption is used to sign tokens.	Symmetric key encryption shall not be used.
Each IM keeps an administration that maps the RU and public key(s) required to verify tokens issued by that RU. This administration is used in the authorisation process in the SFERA service.	Refer to Key exchange procedures are not specified by this IRS. for detailed specification.
IM-RU setup: the RU is responsible for the token issuing process	
IM-Train setup: the RU is responsible for the token issuing process	
AoE setup: the IM is responsible for setting up a token issuing process that maps the authorisations obtained through the ETCS processes to SFERA authorisation tokens	
Basic authorisation to the SFERA service is obtained by sending a valid token for each connection setup with the SFERA service. A connection is established if and only if the token is valid and issued by a known RU (token issuer).	Basic authorisation.
With every publish and subscribe operation, a valid token shall be sent. Publish and subscribe operations are limited to the train movement scope as indicated by the claims in the token. Upon violation of an au-thorisation, an IM shall close the connection.	Fine-grained authorisa-tion
An RU authorises all activities performed by the driv-ers it is responsible for, even if the train movement involves multiple IMs and multiple countries.	An IM should have agreements with all RUs it accepts on its network, including foreign RUs, and should accept tokens for all these RUs
Tokens should be issued for relatively short validity periods and be re-issued (refreshed) frequently.	Typical token validity is an hour.
The IM shall refuse connection requests not bearing a token at all or not bearing a valid token.	
The IM shall close a connection with a DAS as soon as the validity of the related token expires and in case of a missing or invalid token.	

NB:

- This IRS does NOT specify the authentication/authorisation mechanism that is used to connect the TMS to the SFERA service;
- This IRS does NOT specify how public key information is exchanged between RUs and IMs (e.g. X509 certificates);
- This IRS does NOT impose device authentication;
- Any OAUTH2-based authorisation procedure can be used to produce the required tokens, but any custom process that produces tokens according to the above specifications will work. The implementation depends on local requirements.

6.4.3 – Security/integrity

The security/integrity specifications are listed in Table 12.

Table 12: Security/integrity design specifications

Design specification	Remark
The MQTT service shall accept only TLS (Transport Layer Security) or TLS over web socket connections.	Http/plain socket con-nections are insecure.
A DAS/RU shall apply server certificate validation	Verify server identity
IM and DAS shall only send data that complies with the referenced XML schema version and shall not ac-cept data that does not comply with the referenced XML schema.	Data integrity
Tokens shall be treated/processed by all parties as confidential information.	

7 – Maintenance of SFERA

See Reference documents - page 6 and Abbreviations and Symbols - page 8.

7.1 – The need for maintenance

There are several internal as well as external reasons for periodically updating the SFERA protocol. IMs or RUs have needs that cannot be fulfilled by the current version of SFERA: new legislations demand adaptations; technological developments or threats (e.g. cyber security) require changes; future international standards (such as TAP/TAF, IFC, EULYNX, RTM) and new versions of AoE SUBSET-126 are published; etc. A procedure shall therefore be provided to cope with these new demands.

7.2 – Procedure to update SFERA

Change requests: All UIC members can send change requests to the UIC by mail, stating the changes they would like to see and giving a contact person for further consultation. The email address will be specified on the UIC website.

First evaluation: The SFERA secretary (i.e. the person mandated by the UIC to coordinate the SFERA activities) will make a first evaluation to establish whether a change request demands a faster response time by the SFERA working group than the regular half-yearly interval. If so, the secretary will consult the chair persons of the SFERA working group and call a meeting.

Periodical update: The SFERA working group will gather twice a year to process all received change requests. The SFERA secretary will prepare the invitation together with the chair persons of the SFERA working group.

Small and big changes: The procedure for updating SFERA seeks to find a balance between speed and meticulousness. To be able to serve the users of SFERA efficiently, the SFERA working group is mandated to publish new versions of SFERA without consulting all UIC members for changes that have limited impact. For changes with a large impact on the structure and functioning of SFERA, a formal UIC publishing procedure will be followed. The working group consists of experts and is therefore entitled to make this distinction themselves.

Publication: All versions of SFERA will be published on the UIC website. It will be possible to find all supported versions there, accompanied by release notes explaining the differences to previous versions. The release notes will include the migration strategy. Normally a minor change (having backwards compatibility with older minor versions) can take effect immediately.

Questions and support: Although there is no formal helpdesk in place to assist UIC members or companies wanting to build solutions using SFERA, the SFERA secretary can assist by getting in touch with UIC members that have already implemented SFERA. An implementation guide with some examples will be available on the web page.

Complaints: If a UIC member has a complaint about a change request being denied or processed in a way that was not expected, the SFERA secretary can be contacted for an explanation.

7.3 – How to make solutions compatible with the latest version of SFERA

The need for upgrading solutions: One of the aims of SFERA is to facilitate interoperability. If an international train runs in three different countries and each IM supports a different major version of SFERA, this goal will not be met. Therefore, for each implementation, IMs and RUs (and their supporting manufacturers) should have procedures in place to deal with upgrades. In this way they will be able to maintain compatibility with the latest version.

Minor and major updates: SFERA makes a distinction between minor updates and major updates:

- A minor update is a non-breaking change with respect to data structure AND semantics;
- A major update is a breaking change, either in data structure, in semantics, in both data structure and semantics or in the messaging and authentication mechanisms.

Version numbers will change with minor upgrades from e.g. Version 2.3 to 2.4 and with major upgrades from e.g. Version 3.2 to 4.0.

Establishing procedures for upgrading: In order to maintain compatibility while upgrading, procedures should be mutually agreed by IMs and RUs. To make life easier, it is highly recommended that solutions are able to support two major versions for a limited period of time. As a basis for these procedures, some rules are provided to maintain reasonable life cycle management processes on both sides (trackside and on board), with a fair split of the workload:

1. SFERA major and minor version numbers are included in any SFERA message that is exchanged;
2. The DAS-OB should be able to deal with all minor versions within the same major version (e.g. it should be able to ignore unknown elements and attributes, and new attributes/elements must be qualified as optional);
3. The DAS-TS should accept all minor versions within the supported major versions;
4. If conversion of data between versions is needed, the DAS-TS should perform it;
5. The DAS-TS is required to support previous major version(s) for an agreed period of time;
6. The DAS-TS is allowed to upgrade to a new minor version at any time;
7. The DAS-OB shall upgrade to a new major version, taking into account the agreed migration period.

These rules are based on assumptions that the DAS-TS most often are single central systems; not really resource-limited; and with a great deal of control over the rollout process of new software. The DAS-OB, on the other hand, can be one of many different types of devices from different owners and manufacturers, with serious resource constraints. There is less control over the software rollout process, at least in terms of timing (rollout to many devices). Breaking changes in the underlying messaging system (MQTT) and in JWT token specifications are very rare: they will require separate service end points at the DAS-TS, and the migration strategy should be considered case by case. Therefore, most SFERA version updates can be done without change to the underlying messaging system. Most likely, the major changes will be changes to the XSD that are not compatible with older versions (e.g. changing optional elements into mandatory ones, or changing choices in an enumerated list).

Appendix A: SFERA Use Cases

This is an external appendix to the IRS, available on the UIC website: https://appendices.uic.org

Appendix B: SFERA Communication Protocol

This is an external appendix to the IRS, available on the UIC website: https://appendices.uic.org

Appendix C: SFERA Implementation Cases

The SFERA implementation cases are a representation of the way the different layers of the SFERA protocol interact with each other in different scenarios. The focus is on the communication layer and the use cases, taking into account the different external events that impact DAS operational use. They also describe the DAS status the DAS-OB, the DAS-TS and the TMS will consider.

The main objective in introducing the implementation cases into the SFERA Group workflow was to assemble the individual layers of the protocol to compile a global image that would help identify inconsistencies between the different aspects of the protocol.

Implementation cases are included in the IRS because they could be useful:

- for people in the process of understanding the way the SFERA protocol works;

- as support for discussions between RUs and IMs when discussing the possible implementation options to be used on a given network.

The implementation cases have only been described from the IM-Train setup perspective, as this setup takes all layers of SFERA into account. In the IM-RU setup, the functional elements should be equivalent from a device and IM DAS-TS point of view (mainly the statuses). The RU DAS-TS would only add a layer that abstracts the statuses of the different devices to give a unique status to the IM DAS-TS.

All the implementation cases can be implemented using the SFERA protocol. However, there is no obligation to implement all of them on a given network or for a given RU. Moreover, the implementation cases are not exhaustive. This is subject to discussion between individual RUs and IMs for the purpose of agreeing on the scope of their specific implementation.

A full list and description of the implementation cases is provided in an external appendix to the IRS, available on the UIC website: https://appendices.uic.org.

Appendix D: SFERA XML Schema Definition

This is an external appendix to the IRS, available on the UIC website: https://appendices.uic.org

Appendix E: SFERA Data Handbook

This is an external appendix to the IRS, available on the UIC website: https://appendices.uic.org

Appendix F: Sample SFERA XML Communications

This is an external appendix to the IRS, available on the UIC website: https://appendices.uic.org

Appendix G: Correspondence Table SUBSET-126/ SFERA

This is an external appendix to the IRS, available on the UIC website: https://appendices.uic.org

Appendix H: Error codes in SFERA

This appendix contains a list of error codes for SFERA messages. The list is provided as an external document (available on the UIC website: https://appendices.uic.org), as it may need to be continuously updated following the requirements surfacing from the SFERA maintenance process (see point 7 - page 53).

Appendix UIC 1 - Regulatory listing

This appendix is an integral part of the following IRS:

IRS-90940:2020,Ed1: Data Exchanges with Driver Advisory Systems: SFERA. Standardising DAS data exchange for interoperability

The table(s) below list(s) the regulatory source(s) specifically citing this IRS of which UIC has been made aware by the relevant bodies. If applicable, users shall check the validity of this table(s) with the source indicated.

Table UIC-1.1: European regulatory sources citing IRS 90940

Source: title	Source: passage	IRS: passage targeted	Remarks
none			

Bibliography

The following documents are recommended for the understanding or for more in-depth information about subjects treated in this IRS:

1. *UIC Leaflet 920-1: Standard numerical coding for railway undertakings, infrastructure managers and other companies involved in rail-transport chains, Obligatory,* 6th edition, November 2005 - Original, 4/05 (issued 1/11/2005), ISBN 2-7461-1023-7

2. *ERTMS/ETCS SUBSET-023: Glossary of Terms and Abbreviations. Version 3.1.0*. https://www.era.europa.eu/sites/default/files/filesystem/ertms/ccs_tsi_annex_a_-_mandatory_specifications/set_of_specifications_2_etcs_b3_mr1_gsm-r_b1/index003_-_subset-023_v310.pdf

3. *SHIFT2RAIL Terminology for the future Signalling and Automation System: Published by X2Rail-1 - Result and Publications - D2.3* https://projects.shift2rail.org/download.aspx?id=2ecc3978-e6bb-4621-99bc-8532d50722eb

4. *RSSB Interface Requirements for Connected Driver Advisory System (C-DAS)* https://catalogues.rssb.co.uk/rgs/standards/RIS-0711-CCS%20Iss%201.pdf

5. *H.S. Hansen et al.: Using operational data to estimate the running resistance of trains. Estimation of the resistance in a set of Norwegian tunnels, Journal of Rail Transport Planning & Management, 2017* https://www.sciencedirect.com/science/article/pii/S2210970616300415#bib21

6. *Deutsche Bahn AG, 2016. First fully connected Driver Advisory System* https://www.db-systemtechnik.de/dbst-en/news/news/pi_esg_driver_advisory_system-stagecoach_group-1679332

7. *Wikipedia: Punktförmige Zugbeeinflussung* https://de.wikipedia.org/wiki/Punktf%C3%B6rmige_Zugbeeinflussung

8. *Wikipedia: Transmission Voie-Machine* https://en.wikipedia.org/wiki/Transmission_Voie-Machine

List of tables

Table 1: Interoperability requirements 25

Table 2: Evolution requirements 27

Table 3: Requirements supporting different setups 27

Table 4: Communications requirements 28

Table 5: Data integrity requirements 29

Table 6: Functional requirements 30

Table 7: Maintainability requirements 31

Table 8: Provider and Receiver roles in different setups 41

Table 9: Design specifications for different setups 49

Table 10: Data exchange design specifications 50

Table 11: Authentication/authorisation design specifications 51

Table 12: Security/integrity design specifications 52

List of figures

Fig. 1 - Steps from data to business outcome 15
Fig. 2 - High-level system overview 17
Fig. 3 - DAS Inner/Outer loop 19
Fig. 4 - Opticonduite, "Standalone DAS" in use at SNCF 20
Fig. 5 - TimTim, "Connected DAS" in use at NS 20
Fig. 6 - Possible C-DAS architectures 22
Fig. 7 - Lines with different ATPs crossing IM borders 25
Fig. 8 - Outcome as a function of the number of data elements 26
Fig. 9 - Journey Profile Model - Information Exchange in SFERA 32
Fig. 10 - Data architecture of a sample train journey 35
Fig. 11 - Connection through RU DAS-TS 36
Fig. 12 - Direct connection between IM and on-board train DAS 37
Fig. 13 - ATO over ETCS connection 38
Fig. 14 - G2B vs B2G in IM-Train Setup 43
Fig. 15 - G2B vs B2G in IM-RU Setup 43
Fig. 16 - Different layers in scope of IRS 90940 46
Fig. 17 - DAS RU/IM Ecosystem 49

Warning

No part of this publication may be copied, reproduced or distributed by any means whatsoever, including electronic, except for private and individual use, without the express permission of the International Union of Railways (UIC). The same applies for translation, adaptation or transformation, arrangement or reproduction by any method or procedure whatsoever. The sole exceptions - noting the author's name and the source - are "analyses and brief quotations justified by the critical, argumentative, educational, scientific or informative nature of the publication into which they are incorporated". (Articles L 122-4 and L122-5 of the French Intellectual Property Code).

© International Union of Railways (UIC) - Paris, 2020

Published by the International Union of Railways (UIC)
16 rue Jean Rey, 75015 Paris - France,
Dépôt Légal

ISBN 978-2-7461-2929-0